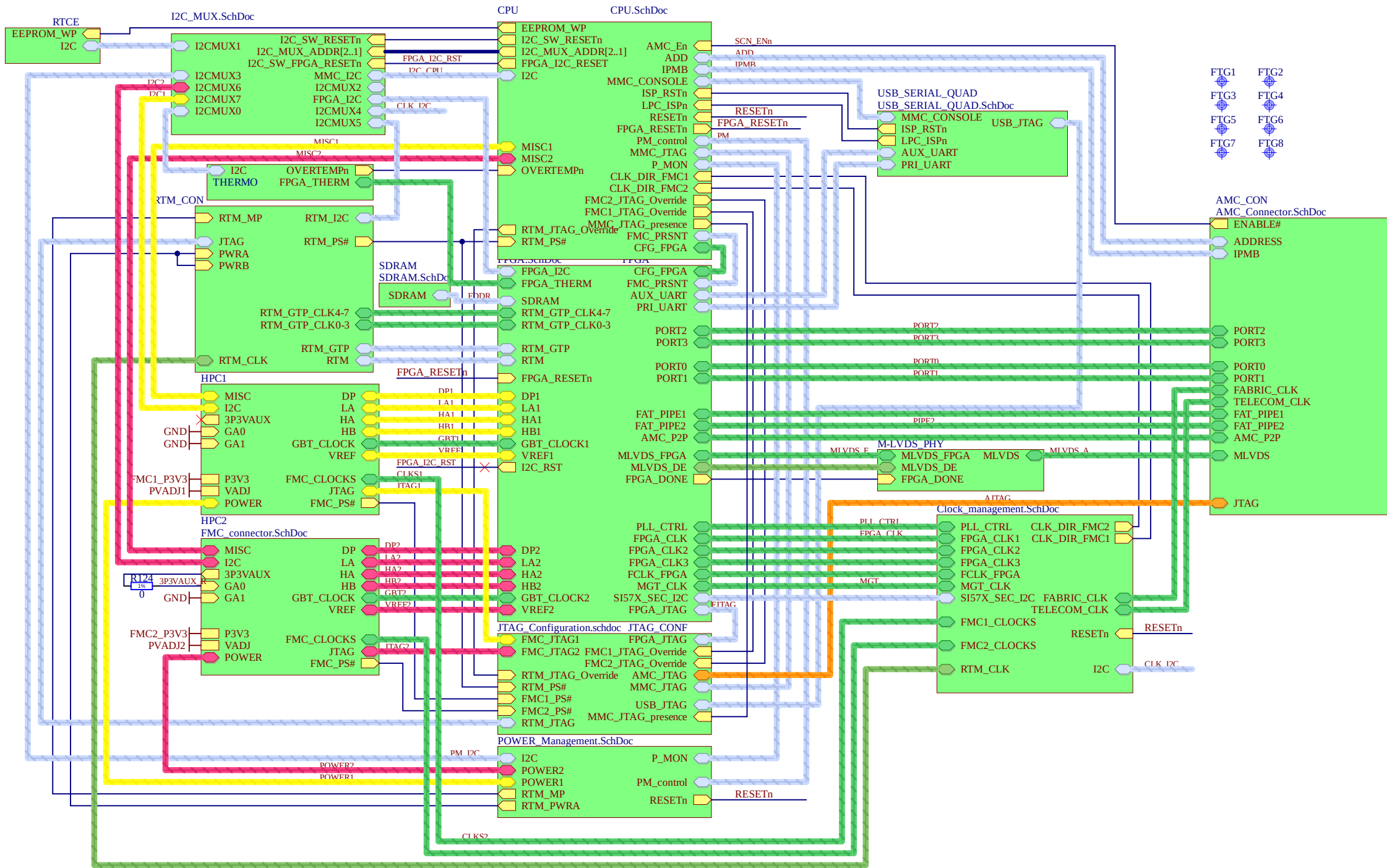
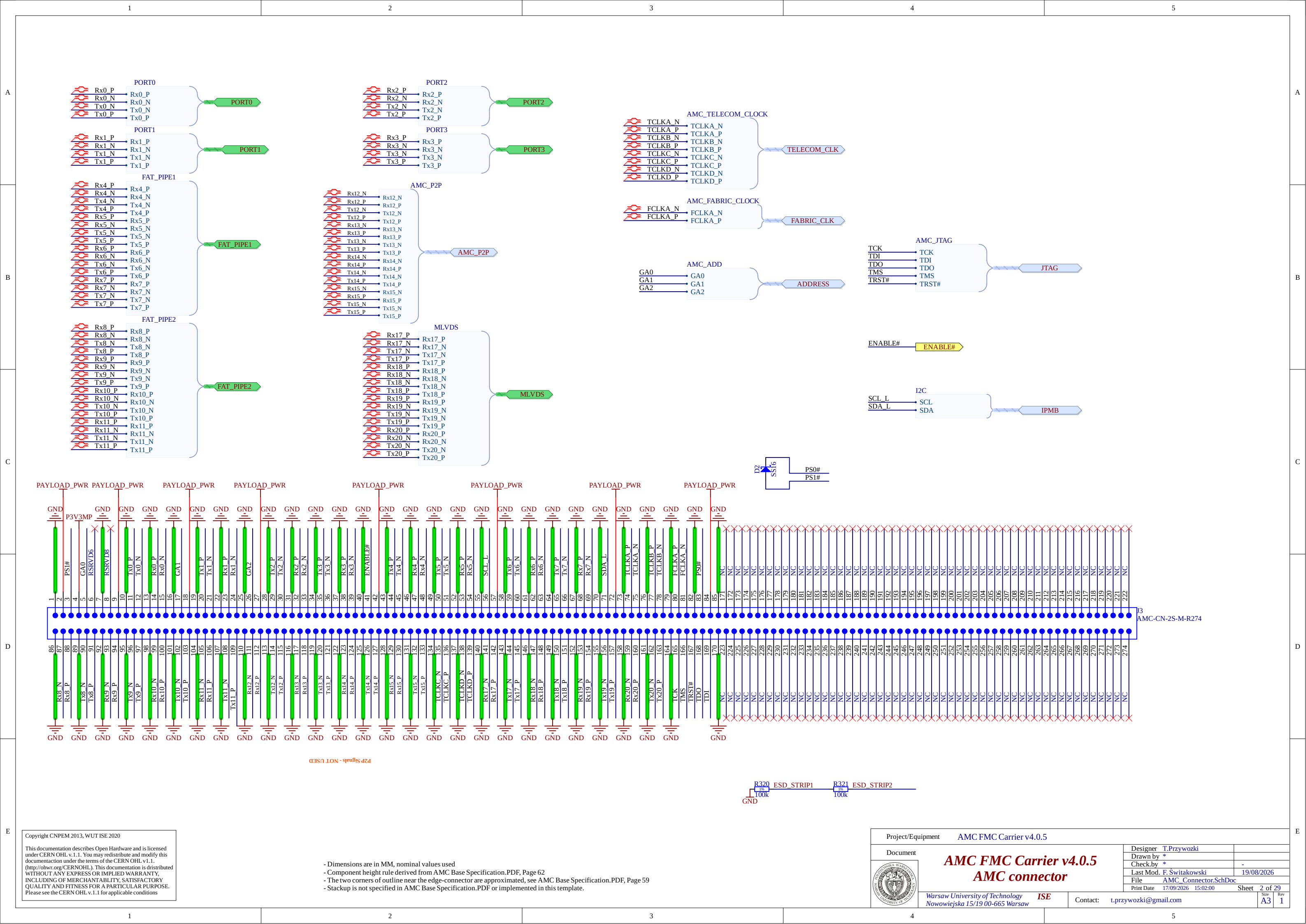


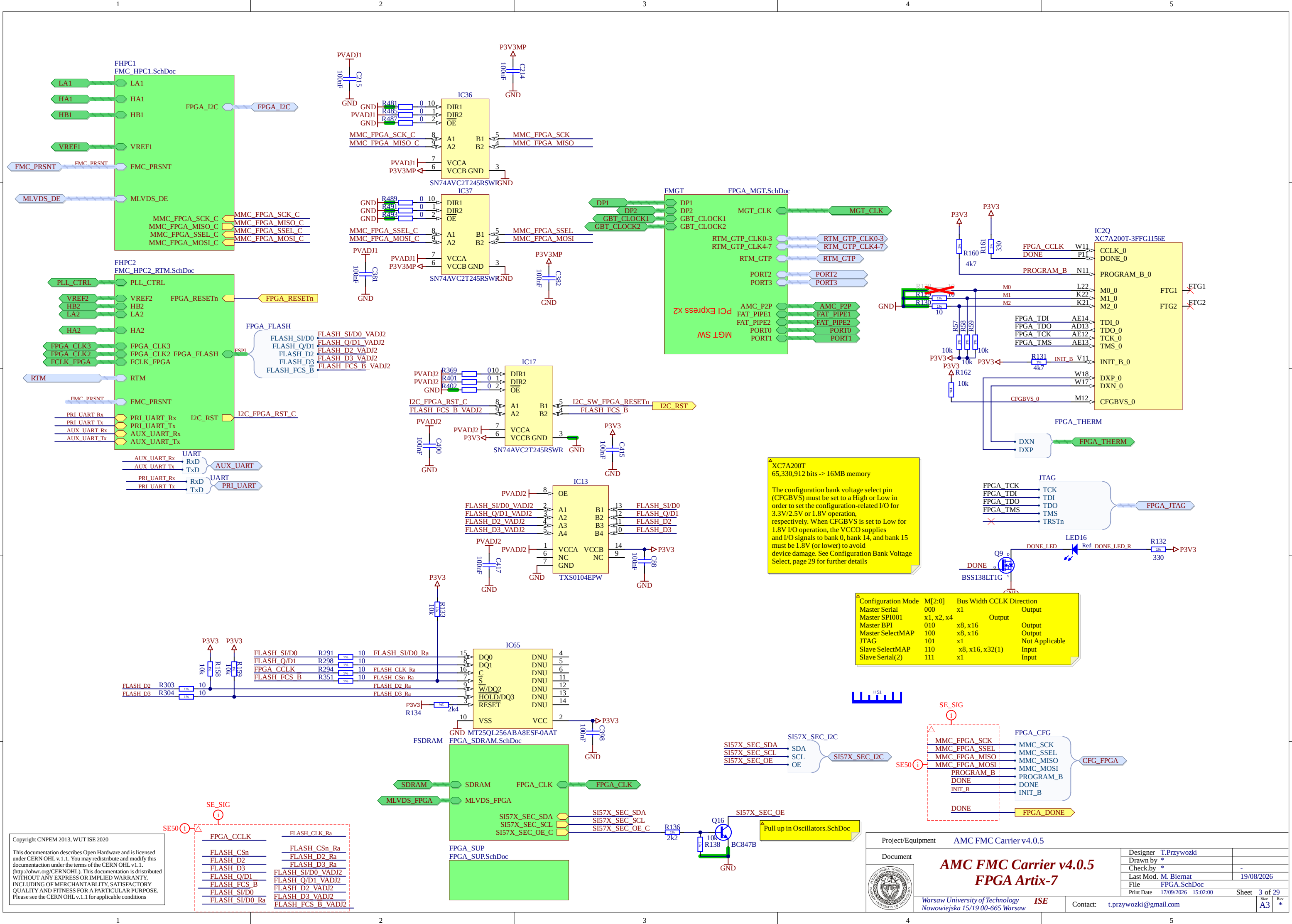


Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

Project/Equipment		AMC FMC Carrier v4.0.5				
Document	AMC FMC Carrier v4.0.5 Top Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw ISE	Designer		T.Przywozki		
		Drawn by		*		
		Check by		*		
		Last Mod.		M. Biernat		
		File		AMC_FMC_Carrier.SchDoc		
Print Date		17/09/2026 15:02:00				
		Sheet	1 of 29			
		Size	A3			
		Rev	1			
Contact:		t.przywozki@gmail.com				

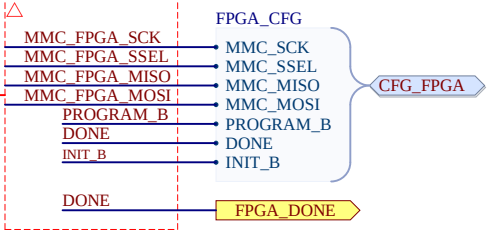




XC7A200T
65,330,912 bits -> 16MB memory

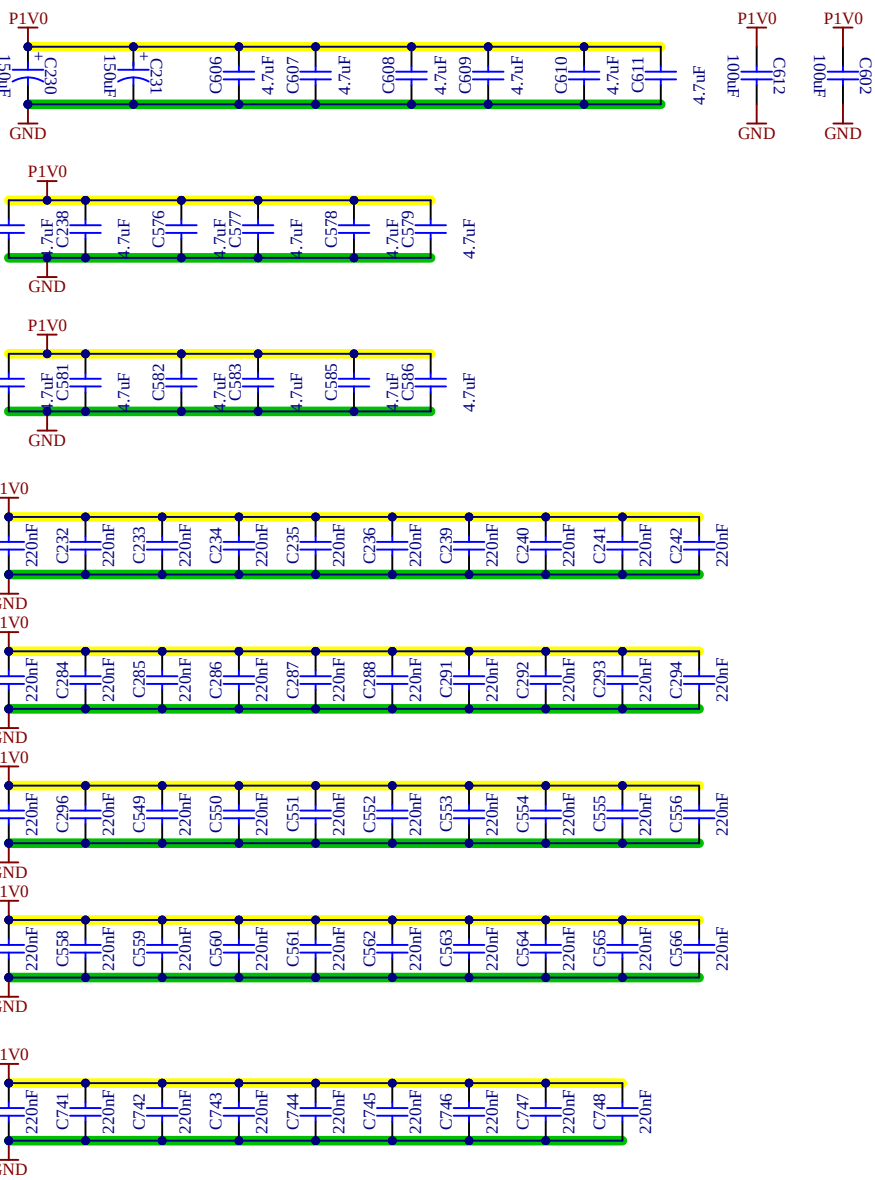
The configuration bank voltage select pin (CFGBVS) must be set to a High or Low in order to set the configuration-related I/O for 3.3V/2.5V or 1.8V operation, respectively. When CFGBVS is set to Low for 1.8V I/O operation, the VCCO supplies and I/O signals to bank 0, bank 14, and bank 15 must be 1.8V (or lower) to avoid device damage. See Configuration Bank Voltage Select, page 29 for further details

Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master Serial	000	x1	Output
Master SPI001	x1, x2, x4		Output
Master BPI	010	x8, x16	Output
Master SelectMAP	100	x8, x16	Output
JTAG	101	x1	Not Applicable
Slave SelectMAP	110	x8, x16, x32(1)	Input
Slave Serial(2)	111	x1	Input



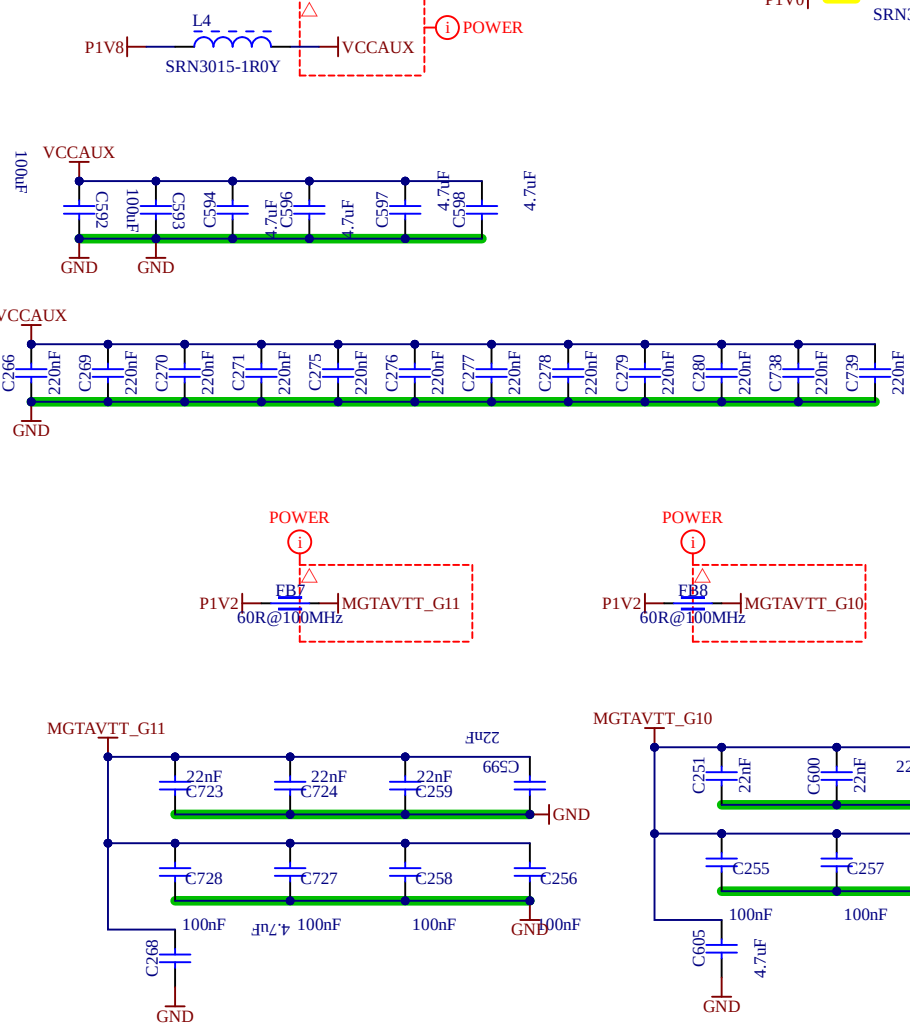
VCCINT

1x680uF, 2x100uF, 28x4.7uF
42x0.47uF on PG.34



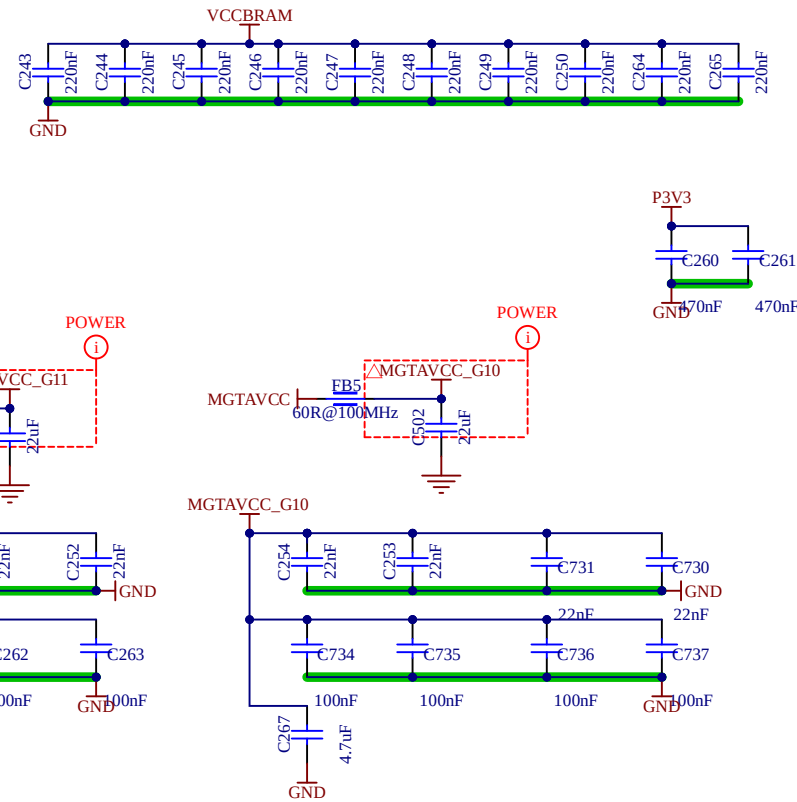
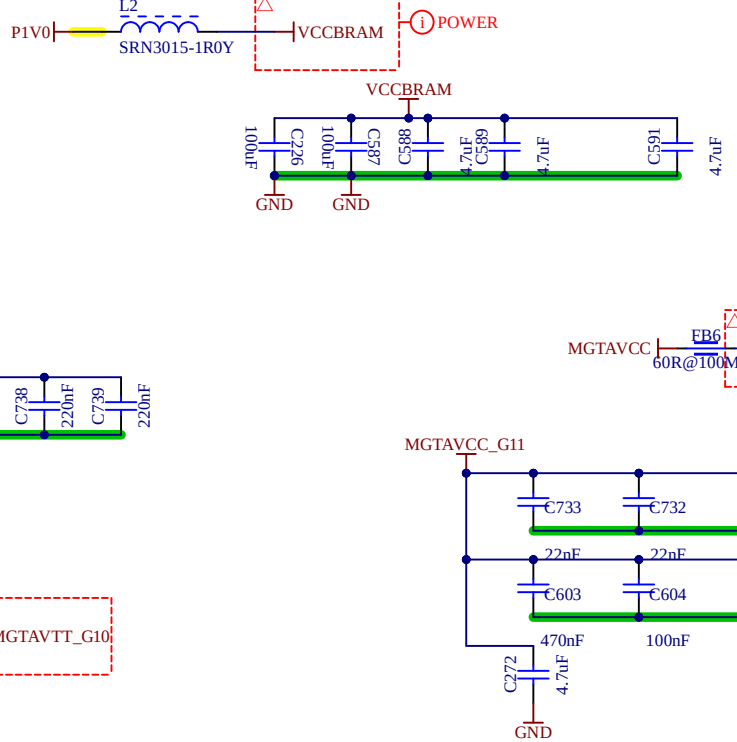
VCCAUX

3x47uF, 4x4.7uF, 7x0.47uF



VCCBRAM

2x100uF, 4x4.7uF, 8x0.47uF



MGTAVCC	1V	2.3A	
MGTAVTT	1.2V	1.5A	
MGTVCCAUX	1.8V	0.5A	
VCCAUX	1.8V	1A	seq1
VCCAUX_IO	1.8/2V	0.1A	seq1
VCCBRAM	1V	1.8A	seq0
VCCINT	1V	6A	seq0
VCCO_1,2	1.2V	3.2A	seq2
VCCO_1,35	1.35V	0.9A	seq2
VCCO_1,5	1.5V	0.9A	seq2
VCCO_1,8	1.8V	0.9A	seq2
VCCO_2,5	2.5V	0.9A	seq2
VCCO_3,3	3.3V	0.9A	seq2

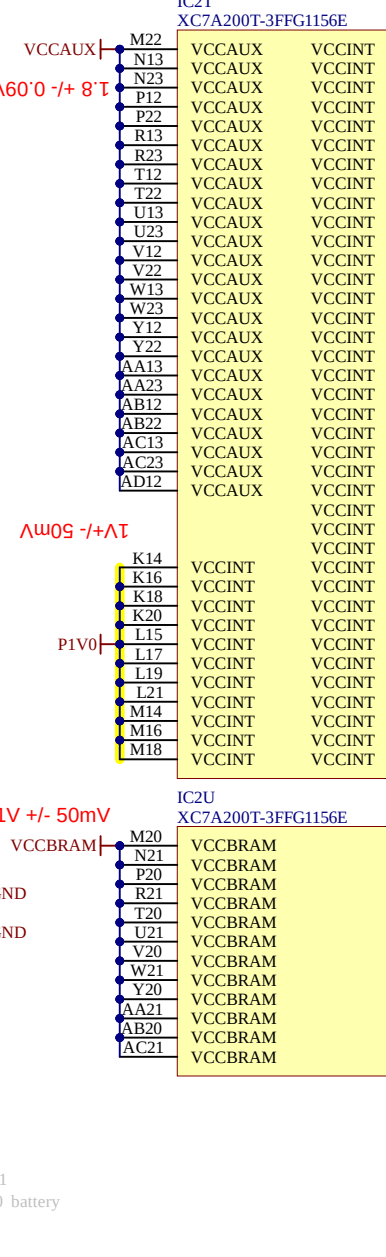
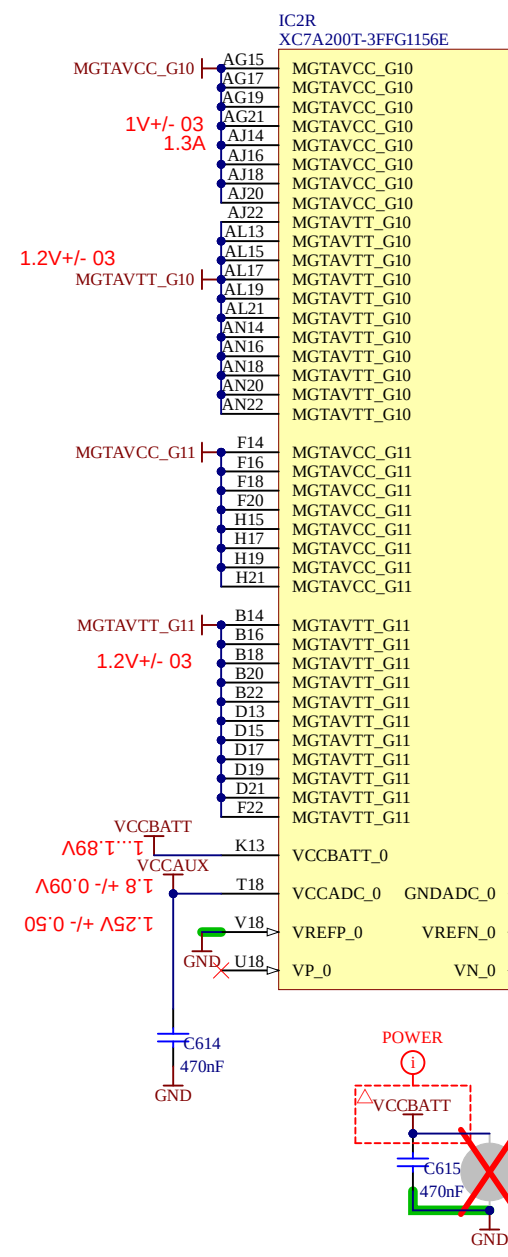
power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO voltage difference between VCCO and VCCAUX must not exceed 2.625V for longer than TVCCO2VCCAUX

power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VMGTAVTT or VMGTAVCC, VCCINT, VMGTAVTT. There is no recommended sequencing for VMGTAVCCAUX. Both VMGTAVCC and VCCINT can be ramped simultaneously. The recommended power-off sequence is the reverse of the power-on sequence to achieve minimum current draw.

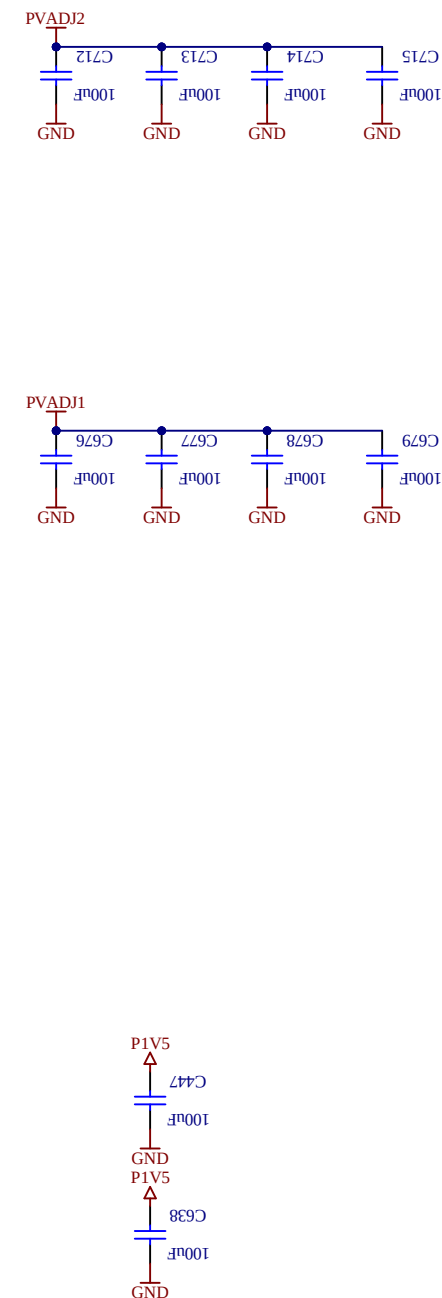
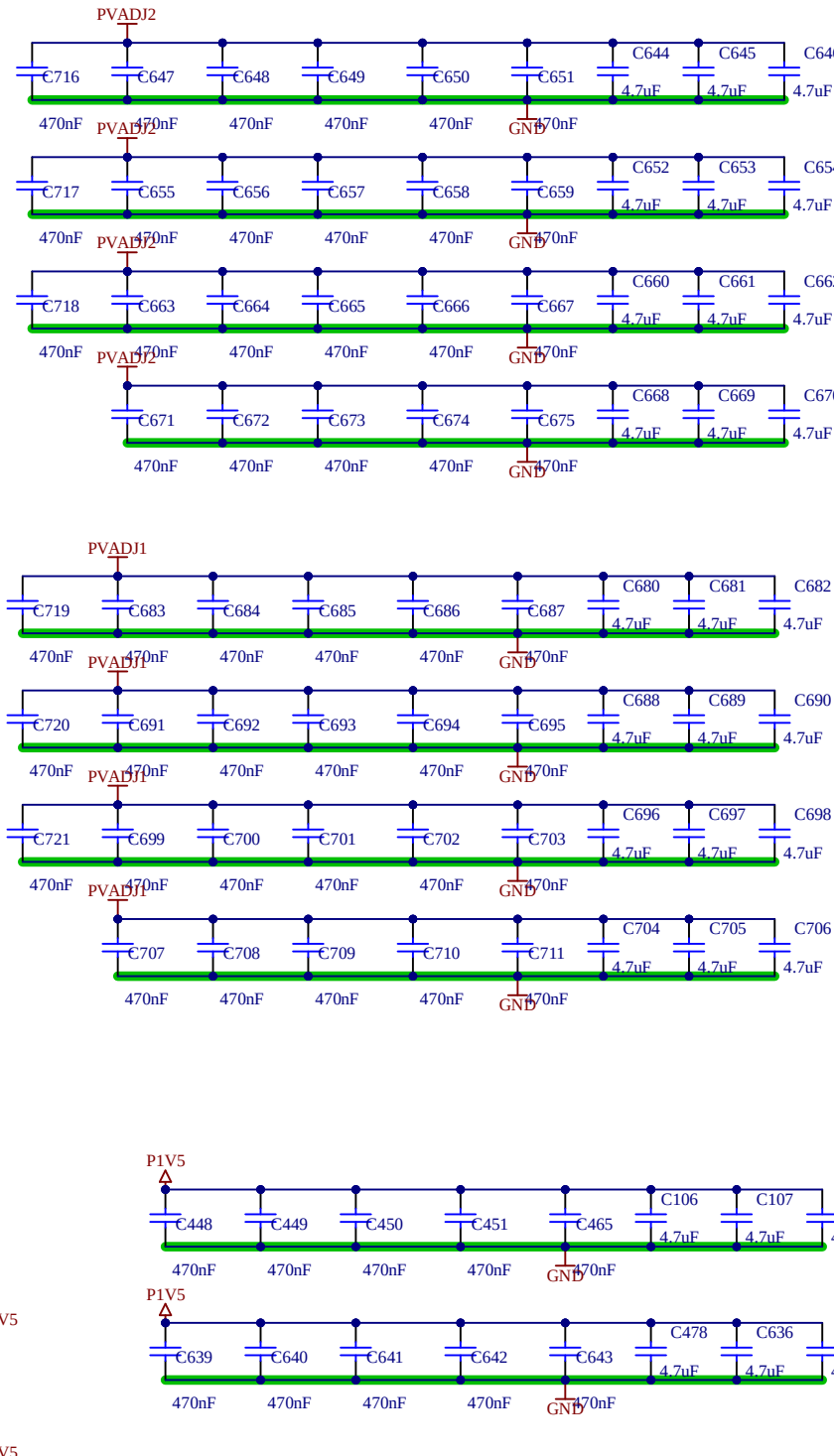
When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT - VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the VMGTAVTT current draw can increase by 460 mA per transceiver during VMGTAVCC ramp up. The duration of the current draw can be up to 0.3 x TMGTAVCC (ramp time from GND to 90% of VMGTAVCC). The reverse is true for power-down.

When VMGTAVTT is powered before VCCINT and VMGTAVTT - VCCINT > 150 mV and VCCINT < 0.7V, the VMGTAVTT current draw can increase by 50 mA per transceiver during VCCINT ramp up. The duration of the current draw can be up to 0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true for power-down.

IC2V XC7A200T-3FFG1156E	IC2W XC7A200T-3FFG1156E	IC2R XC7A200T-3FFG1156E	IC2T XC7A200T-3FFG1156E	IC2S XC7A200T-3FFG1156E	IC2U XC7A200T-3FFG1156E
A11 GND	V13 GND	AG15 MGTAVCC_G10	M22 VCCAUX	L13 VCCO_0	M20 VCCBRAM
A12 GND	V19 GND	AG17 MGTAVCC_G10	N23 VCCAUX	L14 VCCO_0	N21 VCCBRAM
A14 GND	V21 GND	AG19 MGTAVCC_G10	N23 VCCAUX	L15 VCCO_0	P20 VCCBRAM
A16 GND	V23 GND	AG21 MGTAVCC_G10	P12 VCCAUX	L16 VCCO_0	P20 VCCBRAM
A18 GND	V25 GND	AG21 MGTAVCC_G10	P22 VCCAUX	L17 VCCO_0	P20 VCCBRAM
A20 GND	V27 GND	AG21 MGTAVCC_G10	R13 VCCAUX	L18 VCCO_0	P20 VCCBRAM
A22 GND	V29 GND	AG21 MGTAVCC_G10	R23 VCCAUX	L19 VCCO_0	P20 VCCBRAM
A24 GND	V31 GND	AG21 MGTAVCC_G10	T12 VCCAUX	L20 VCCO_0	P20 VCCBRAM
A26 GND	V33 GND	AG21 MGTAVCC_G10	T22 VCCAUX	L21 VCCO_0	P20 VCCBRAM
B8 GND	V35 GND	AG21 MGTAVCC_G10	U13 VCCAUX	L22 VCCO_0	P20 VCCBRAM
B12 GND	V37 GND	AG21 MGTAVCC_G10	U23 VCCAUX	L23 VCCO_0	P20 VCCBRAM
B33 GND	V39 GND	AG21 MGTAVCC_G10	V12 VCCAUX	L24 VCCO_0	P20 VCCBRAM
C5 GND	V41 GND	AG21 MGTAVCC_G10	V22 VCCAUX	L25 VCCO_0	P20 VCCBRAM
C12 GND	V43 GND	AG21 MGTAVCC_G10	W13 VCCAUX	L26 VCCO_0	P20 VCCBRAM
C13 GND	V45 GND	AG21 MGTAVCC_G10	W23 VCCAUX	L27 VCCO_0	P20 VCCBRAM
C15 GND	V47 GND	AG21 MGTAVCC_G10	Y12 VCCAUX	L28 VCCO_0	P20 VCCBRAM
C17 GND	V49 GND	AG21 MGTAVCC_G10	Y22 VCCAUX	L29 VCCO_0	P20 VCCBRAM
C19 GND	V51 GND	AG21 MGTAVCC_G10	AA13 VCCAUX	L30 VCCO_0	P20 VCCBRAM
C21 GND	V53 GND	AG21 MGTAVCC_G10	AA23 VCCAUX	L31 VCCO_0	P20 VCCBRAM
C23 GND	V55 GND	AG21 MGTAVCC_G10	AB12 VCCAUX	L32 VCCO_0	P20 VCCBRAM
C24 GND	V57 GND	AG21 MGTAVCC_G10	AB22 VCCAUX	L33 VCCO_0	P20 VCCBRAM
C30 GND	V59 GND	AG21 MGTAVCC_G10	AC13 VCCAUX	L34 VCCO_0	P20 VCCBRAM
D2 GND	V61 GND	AG21 MGTAVCC_G10	AC23 VCCAUX	L35 VCCO_0	P20 VCCBRAM
D12 GND	V63 GND	AG21 MGTAVCC_G10	AD12 VCCAUX	L36 VCCO_0	P20 VCCBRAM
D13 GND	V65 GND	AG21 MGTAVCC_G10	K14 VCCINT	L37 VCCO_0	P20 VCCBRAM
D27 GND	V67 GND	AG21 MGTAVCC_G10	K18 VCCINT	L38 VCCO_0	P20 VCCBRAM
E9 GND	V69 GND	AG21 MGTAVCC_G10	K20 VCCINT	L39 VCCO_0	P20 VCCBRAM
E12 GND	V71 GND	AG21 MGTAVCC_G10	L15 VCCINT	L40 VCCO_0	P20 VCCBRAM
E14 GND	V73 GND	AG21 MGTAVCC_G10	L17 VCCINT	L41 VCCO_0	P20 VCCBRAM
E16 GND	V75 GND	AG21 MGTAVCC_G10	L19 VCCINT	L42 VCCO_0	P20 VCCBRAM
E18 GND	V77 GND	AG21 MGTAVCC_G10	L21 VCCINT	L43 VCCO_0	P20 VCCBRAM
E20 GND	V79 GND	AG21 MGTAVCC_G10	L23 VCCINT	L44 VCCO_0	P20 VCCBRAM
E22 GND	V81 GND	AG21 MGTAVCC_G10	L25 VCCINT	L45 VCCO_0	P20 VCCBRAM
E23 GND	V83 GND	AG21 MGTAVCC_G10	L27 VCCINT	L46 VCCO_0	P20 VCCBRAM
E24 GND	V85 GND	AG21 MGTAVCC_G10	L29 VCCINT	L47 VCCO_0	P20 VCCBRAM
F3 GND	V87 GND	AG21 MGTAVCC_G10	M16 VCCINT	L48 VCCO_0	P20 VCCBRAM
F12 GND	V89 GND	AG21 MGTAVCC_G10	M18 VCCINT	L49 VCCO_0	P20 VCCBRAM
F23 GND	V91 GND	AG21 MGTAVCC_G10	N15 VCCINT	L50 VCCO_0	P20 VCCBRAM
F31 GND	V93 GND	AG21 MGTAVCC_G10		L51 VCCO_0	P20 VCCBRAM
G3 GND	V95 GND	AG21 MGTAVCC_G10		L52 VCCO_0	P20 VCCBRAM
G12 GND	V97 GND	AG21 MGTAVCC_G10		L53 VCCO_0	P20 VCCBRAM
G13 GND	V99 GND	AG21 MGTAVCC_G10		L54 VCCO_0	P20 VCCBRAM
G15 GND	V101 GND	AG21 MGTAVCC_G10		L55 VCCO_0	P20 VCCBRAM
G17 GND	V103 GND	AG21 MGTAVCC_G10		L56 VCCO_0	P20 VCCBRAM
G19 GND	V105 GND	AG21 MGTAVCC_G10		L57 VCCO_0	P20 VCCBRAM
G21 GND	V107 GND	AG21 MGTAVCC_G10		L58 VCCO_0	P20 VCCBRAM
G23 GND	V109 GND	AG21 MGTAVCC_G10		L59 VCCO_0	P20 VCCBRAM
G28 GND	V111 GND	AG21 MGTAVCC_G10		L60 VCCO_0	P20 VCCBRAM
H10 GND	V113 GND	AG21 MGTAVCC_G10		L61 VCCO_0	P20 VCCBRAM
H13 GND	V115 GND	AG21 MGTAVCC_G10		L62 VCCO_0	P20 VCCBRAM
H22 GND	V117 GND	AG21 MGTAVCC_G10		L63 VCCO_0	P20 VCCBRAM
H23 GND	V119 GND	AG21 MGTAVCC_G10		L64 VCCO_0	P20 VCCBRAM
H25 GND	V121 GND	AG21 MGTAVCC_G10		L65 VCCO_0	P20 VCCBRAM
J7 GND	V123 GND	AG21 MGTAVCC_G10		L66 VCCO_0	P20 VCCBRAM
J14 GND	V125 GND	AG21 MGTAVCC_G10		L67 VCCO_0	P20 VCCBRAM
J15 GND	V127 GND	AG21 MGTAVCC_G10		L68 VCCO_0	P20 VCCBRAM
J16 GND	V129 GND	AG21 MGTAVCC_G10		L69 VCCO_0	P20 VCCBRAM
J17 GND	V131 GND	AG21 MGTAVCC_G10		L70 VCCO_0	P20 VCCBRAM
J18 GND	V133 GND	AG21 MGTAVCC_G10		L71 VCCO_0	P20 VCCBRAM
J19 GND	V135 GND	AG21 MGTAVCC_G10		L72 VCCO_0	P20 VCCBRAM



For VCCO_17 and 12 please follow:
VCCO pins in unbonded banks must be connected to the VCCO for that bank for package migration. Do NOT connect unbonded VCCO pins to different supplies. Without package migration requirement, VCCO pins in unbonded banks can be tied to a common supply (VCCO or ground).
No migration available (since there is no 350T) - I recommend connecting to empty power rail and using as power decoupling placeholder for decaps. Remove banks C22 and C29



Copyright CNPEM 2013, WUTISE 2020
This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (http://ohw.org/cernohl). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

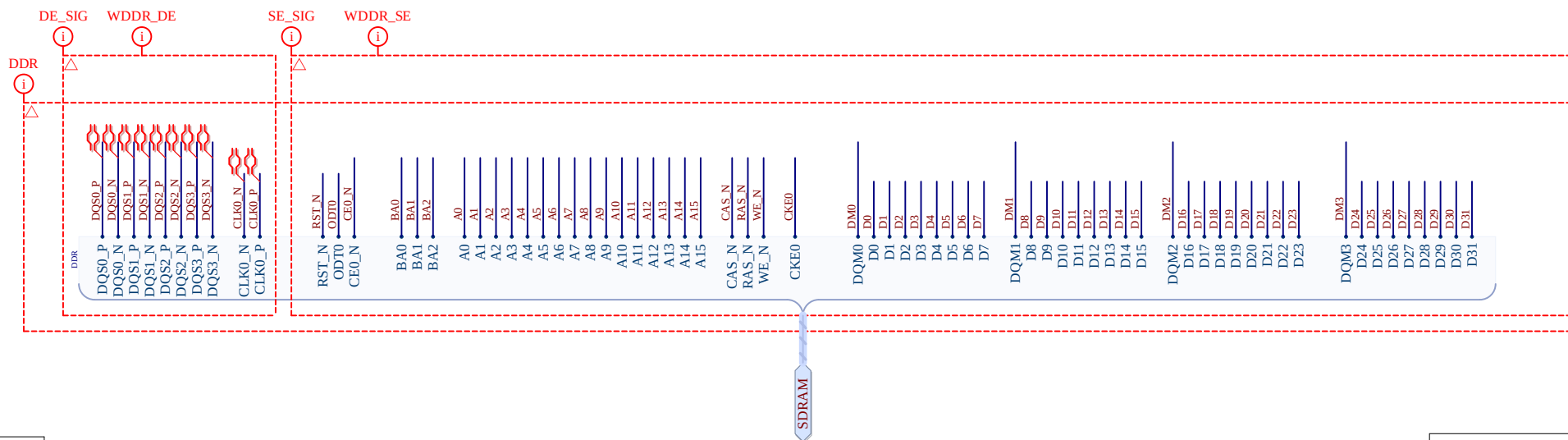
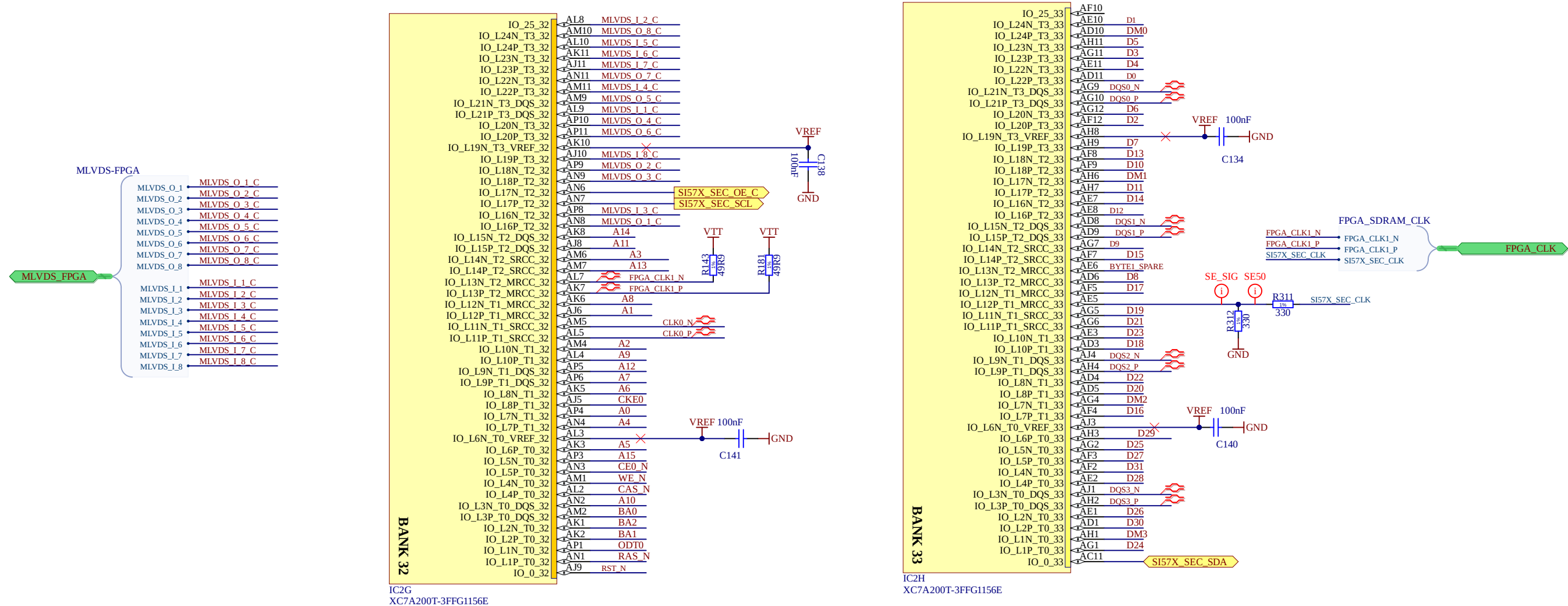
Project/Equipment	AMC FMC Carrier v4.0.5	Designer	T.Przywozki
Document		Drawn by	*
		Check by	*
		Last Mod.	F. Switkowski
		File	FPGA_SUPSchDoc
		Print Date	17/09/2026 15:02:01
		Sheet	4 of 29
		A2	1

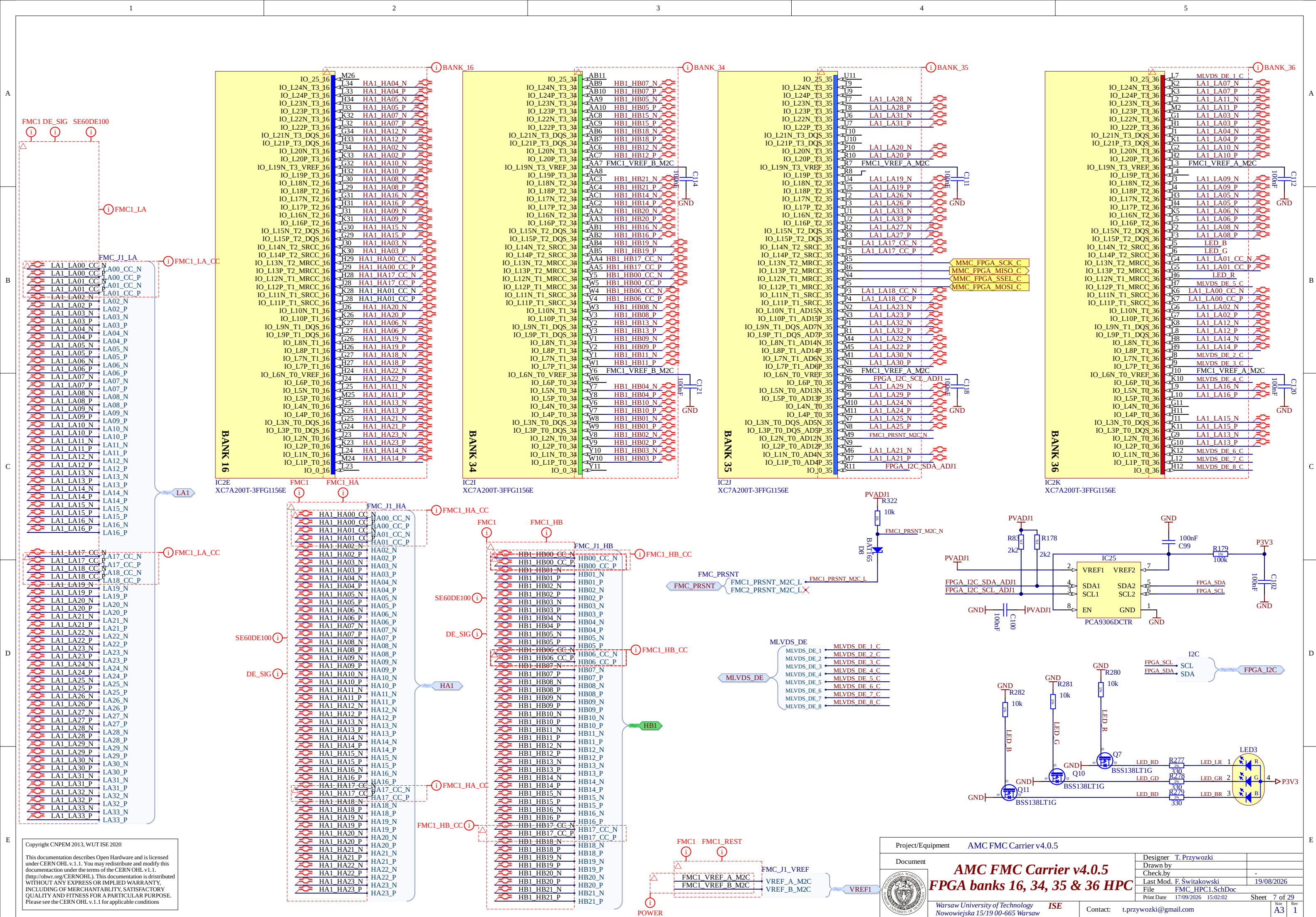


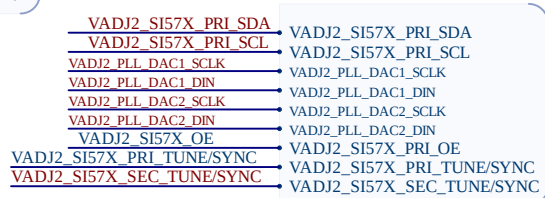
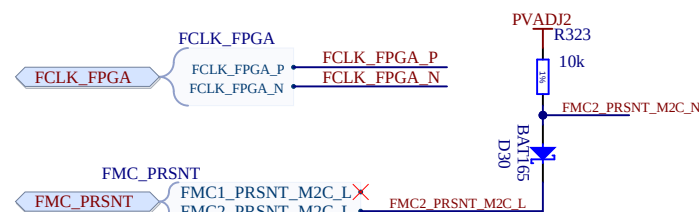
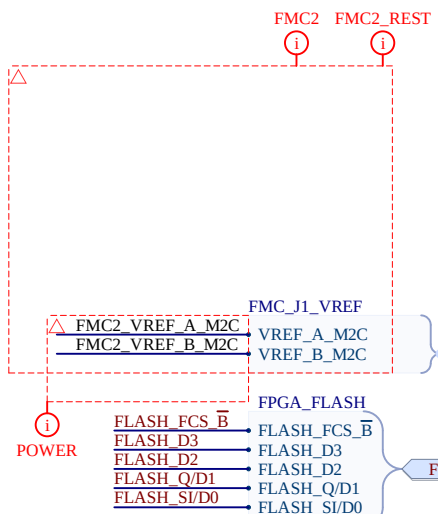
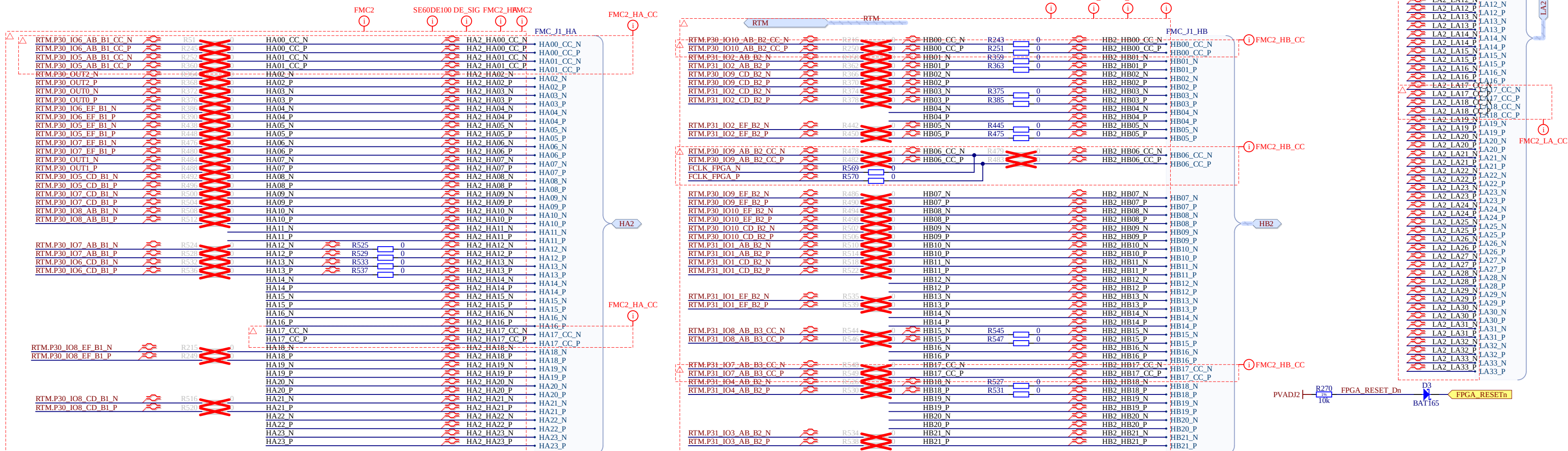
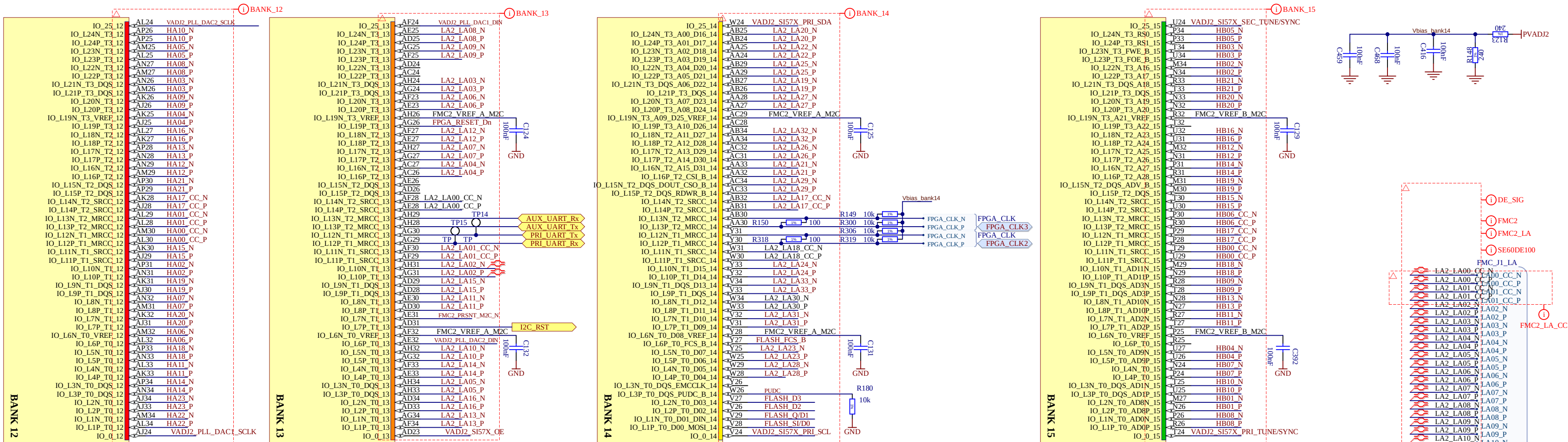
Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

ISE

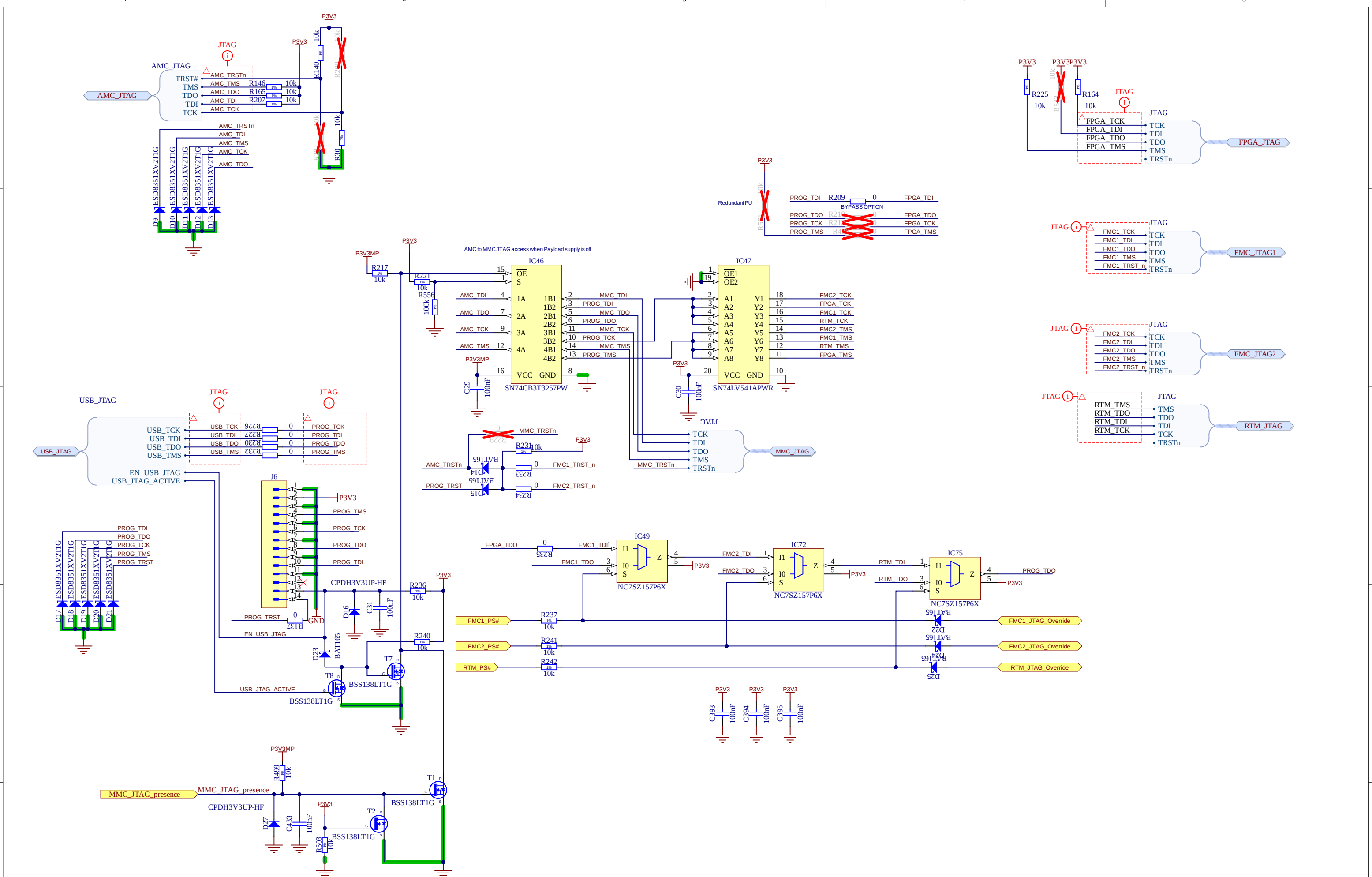
Contact: t.przywozki@gmail.com



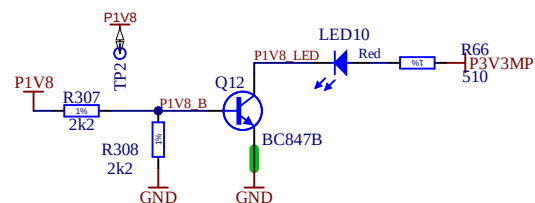
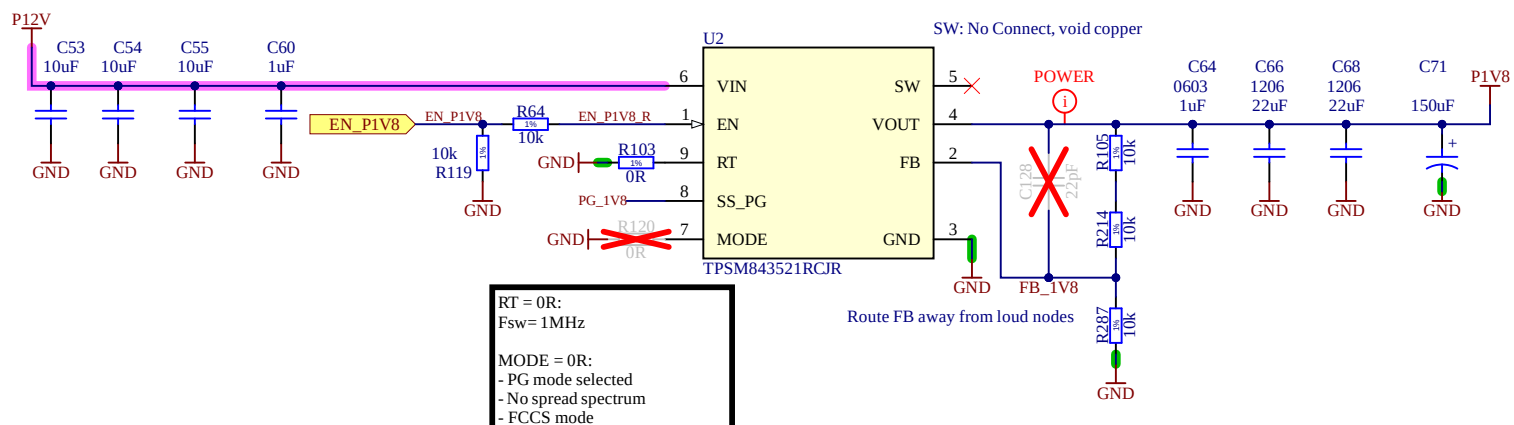
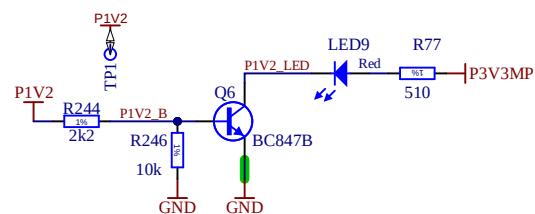
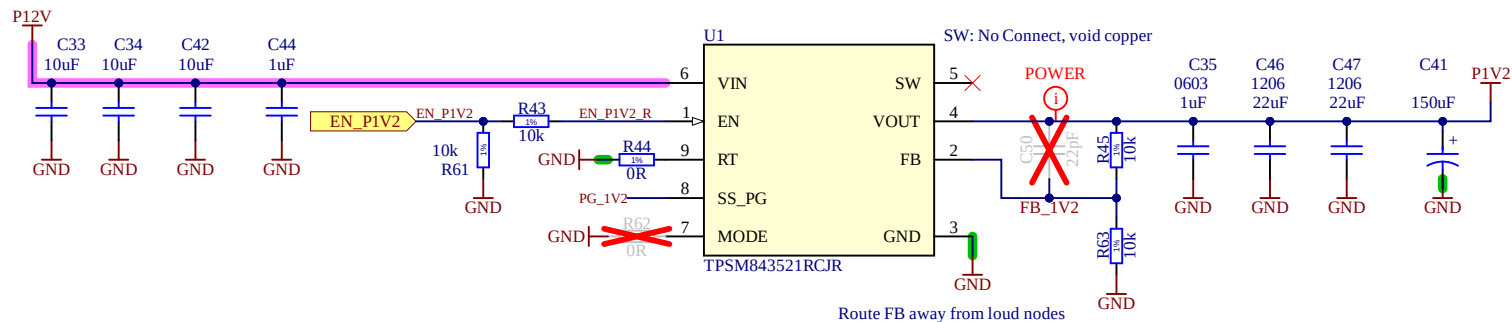




Copyright CNPEM 2013, WUTISE 2020
This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (http://ohw.org/CERN/OHL). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

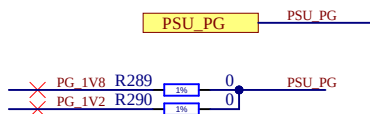


Compatible parts:
TPSM843521 - 5A
TPSM843321 - 3A
TPSM84538 - 5A, 28V in
TPSM84338 - 3A, 28V in



Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

Project/Equipment **AMC FMC Carrier v4.0.5**

Document



AMC FMC Carrier v4.0.5

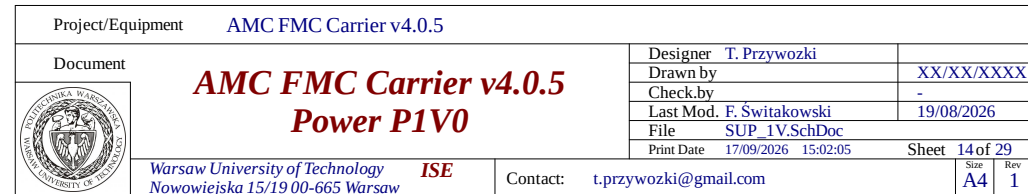
Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

ISE

Contact: t.przywozki@gmail.com

Designer	T. Przywozki		
Drawn by			XX/XX/XXXX
Check.by			-
Last Mod.	F. Świtakowski		19/08/2026
File	SUP_1.2_1.8.SchDoc		
Print Date	17/09/2026	15:02:04	Sheet 11 of 29

Size	Rev
A4	1



This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (<http://ohw.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

AMC FMC Carrier v4.0.5
Power P1V0



ISE

Designer	T. Przywozki		
Drawn by			XX/XX/XXXX
Check by			-
Last Mod.	F. Świtakowski		19/08/2026
File	SUP_1V.SchDoc		
Print Date	17/09/2026	15:02:05	Sheet 14 of 29

Sheet 14 of 29

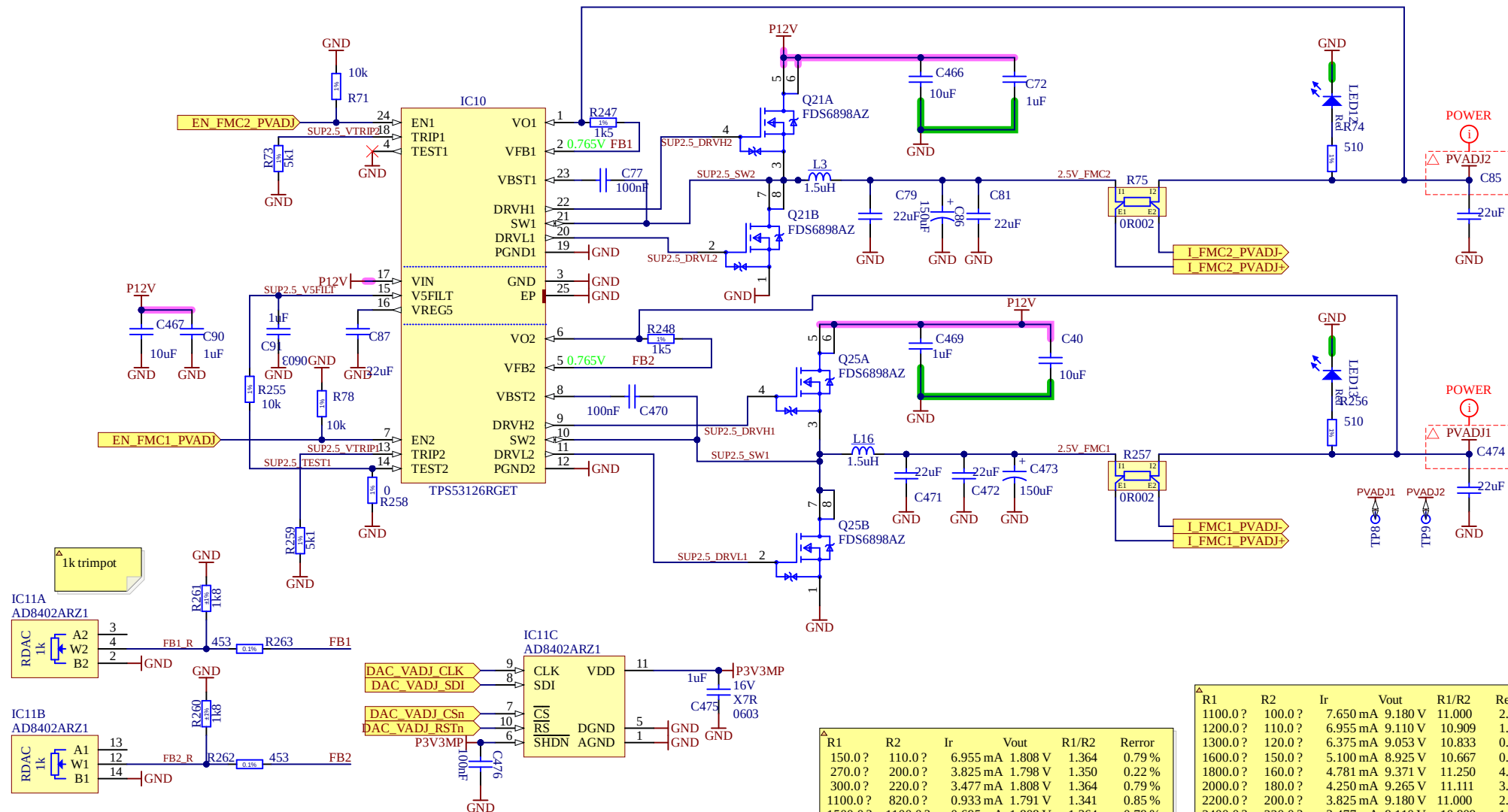
Size	Rev
A 4	1

A

B

C

D



R1	R2	Ir	Vout	R1/R2	Error
1100.0?	100.0?	7.650 mA	9.180 V	11.000	2.19 %
1200.0?	110.0?	6.955 mA	9.110 V	10.909	1.34 %
1300.0?	120.0?	6.375 mA	9.053 V	10.833	0.64 %
1600.0?	150.0?	5.100 mA	8.925 V	10.667	0.91 %
1800.0?	160.0?	4.781 mA	9.371 V	11.250	4.51 %
2000.0?	180.0?	4.250 mA	9.265 V	11.111	3.22 %
2200.0?	200.0?	3.825 mA	9.180 V	11.000	2.19 %
2400.0?	220.0?	3.477 mA	9.110 V	10.909	1.34 %
2700.0?	240.0?	3.188 mA	9.371 V	11.250	4.51 %
3000.0?	270.0?	2.833 mA	9.265 V	11.111	3.22 %
3300.0?	300.0?	2.550 mA	9.180 V	11.000	2.19 %

Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1.1. (<http://ohw.org/CERN/OHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

Project/Equipment AMC FMC Carrier v4.0.5

Document



AMC FMC Carrier v4.0.5 Power FMCs PVADJ

Warsaw University of Technology
Nowowiejska 15/19 00-665 Warsaw

ISE

Contact: t.przywozki@gmail.com

Designer	T. Przywozki	XX/XX/XXXX
Drawn by		-
Check by		-
Last Mod.	F. Świątkowski	19/08/2026
File	SUP_2.5_FMC.SchDoc	
Print Date	17/09/2026 15:02:05	Sheet 15 of 29

Size A4
Rev 1

A

B

C

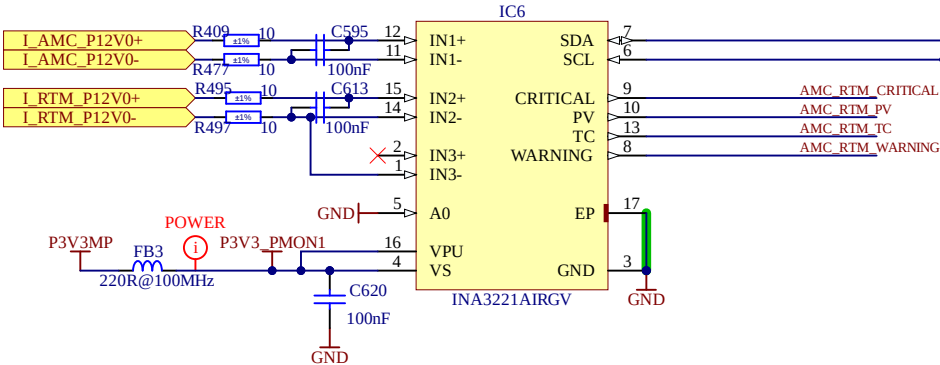
D

A

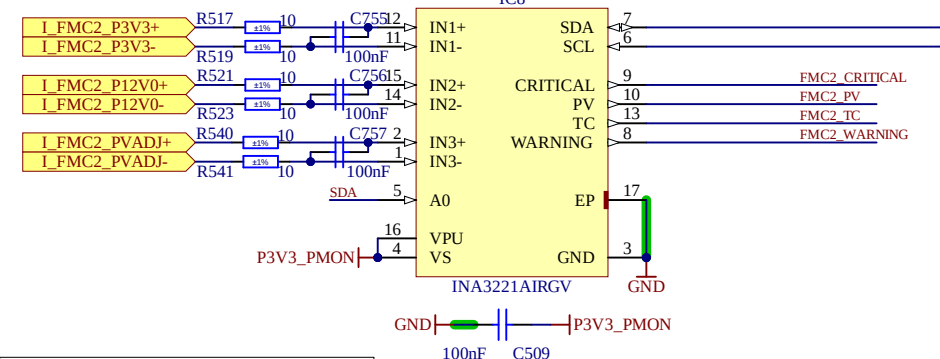
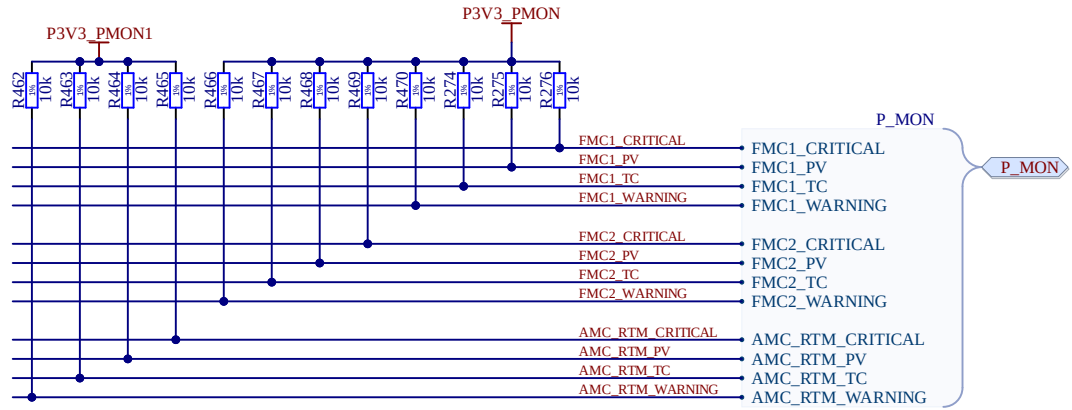
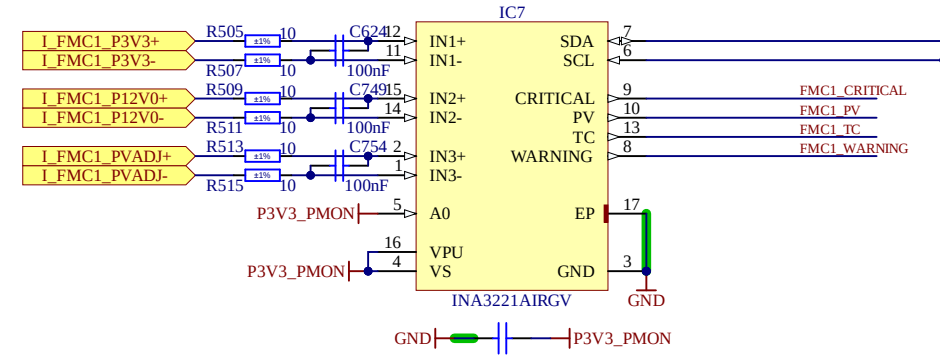
B

C

D



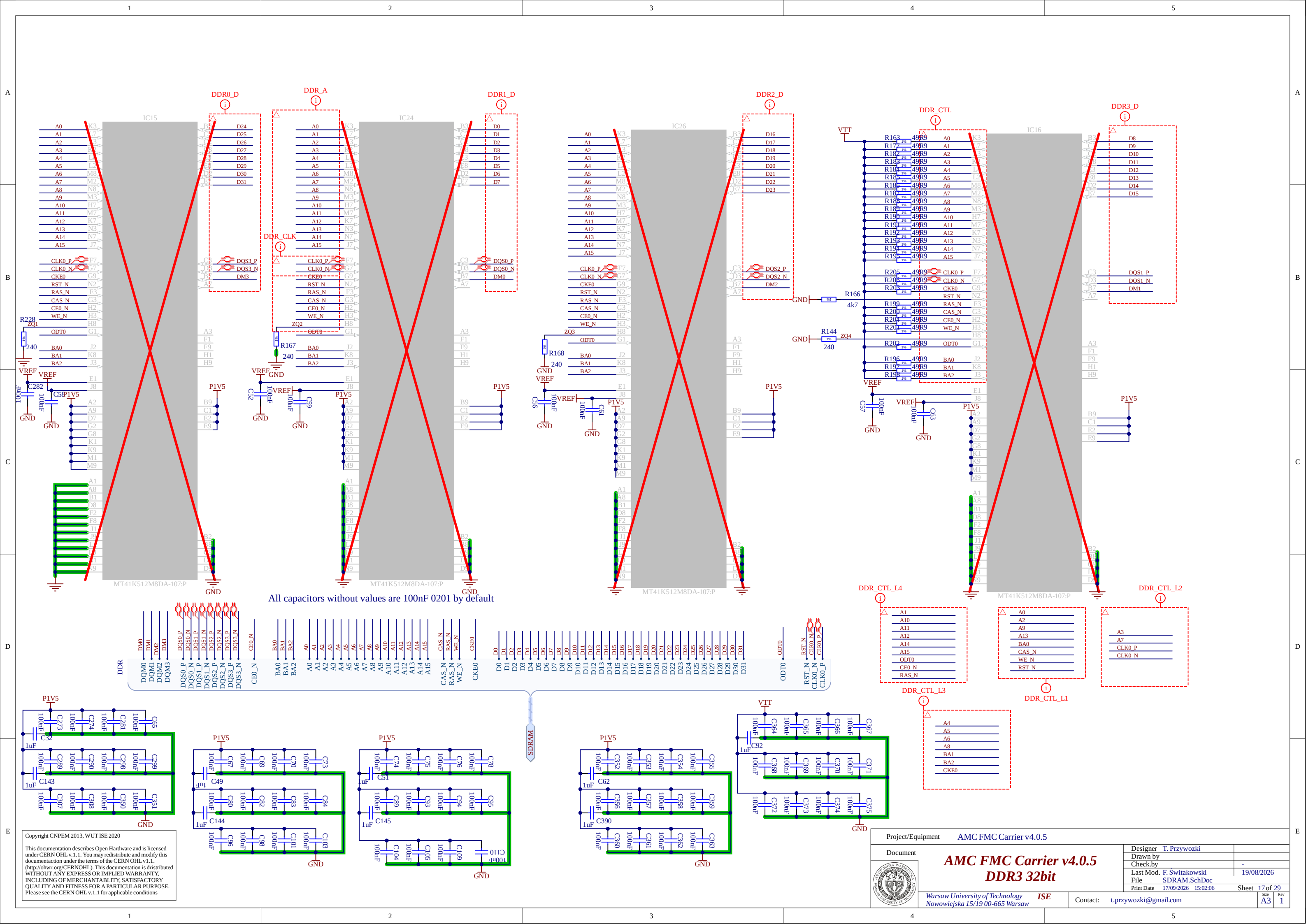
A1	SLAVE ADDRESS
GND	1000000
VS	1000001
SDA	1000010
SCL	1000011

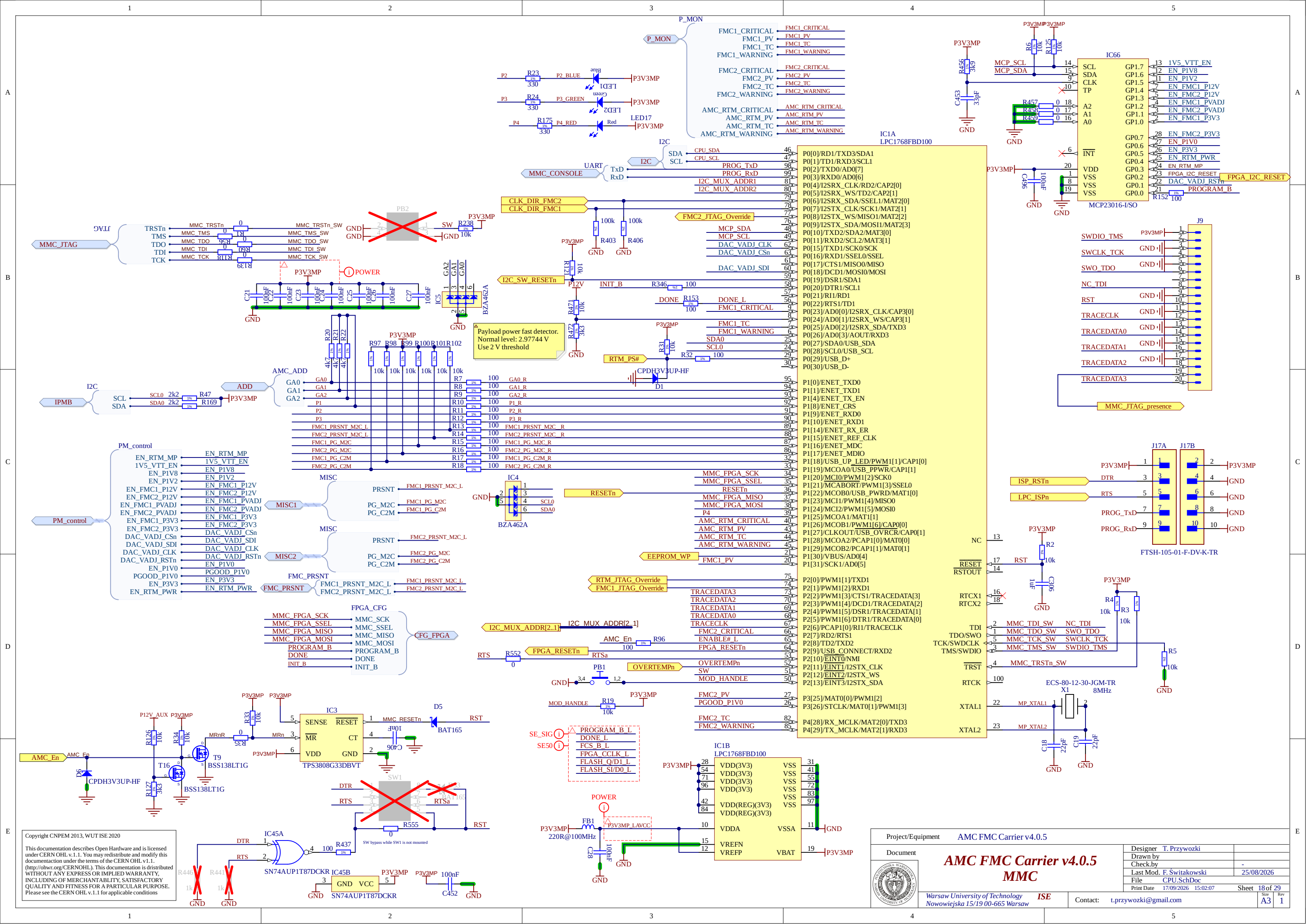


Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

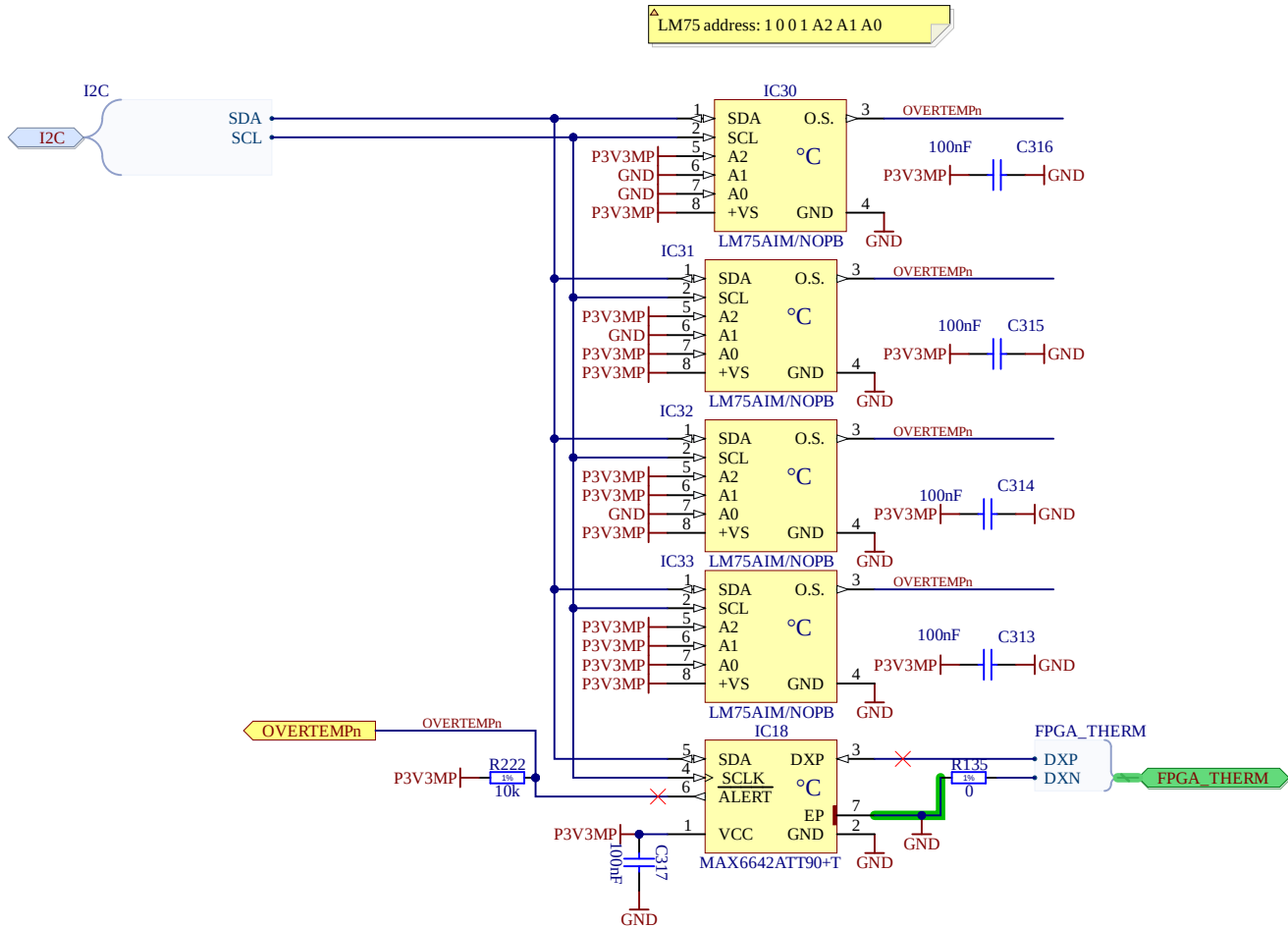
Project/Equipment		AMC FMC Carrier v4.0.5		
Document		AMC FMC Carrier v4.0.5 Power Measurement		
	Designer	T. Przywozki		
	Drawn by		XX/XX/XXXX	
	Check by		-	
	Last Mod.	F. Świtakowski	19/08/2026	
	File	UI_mon.SchDoc		
Print Date		17/09/2026 15:02:06	Sheet 16 of 29	
Contact:		t.przywozki@gmail.com	Size A4 Rev 1	





Copyright CNPEM 2013, WUT ISE 2020

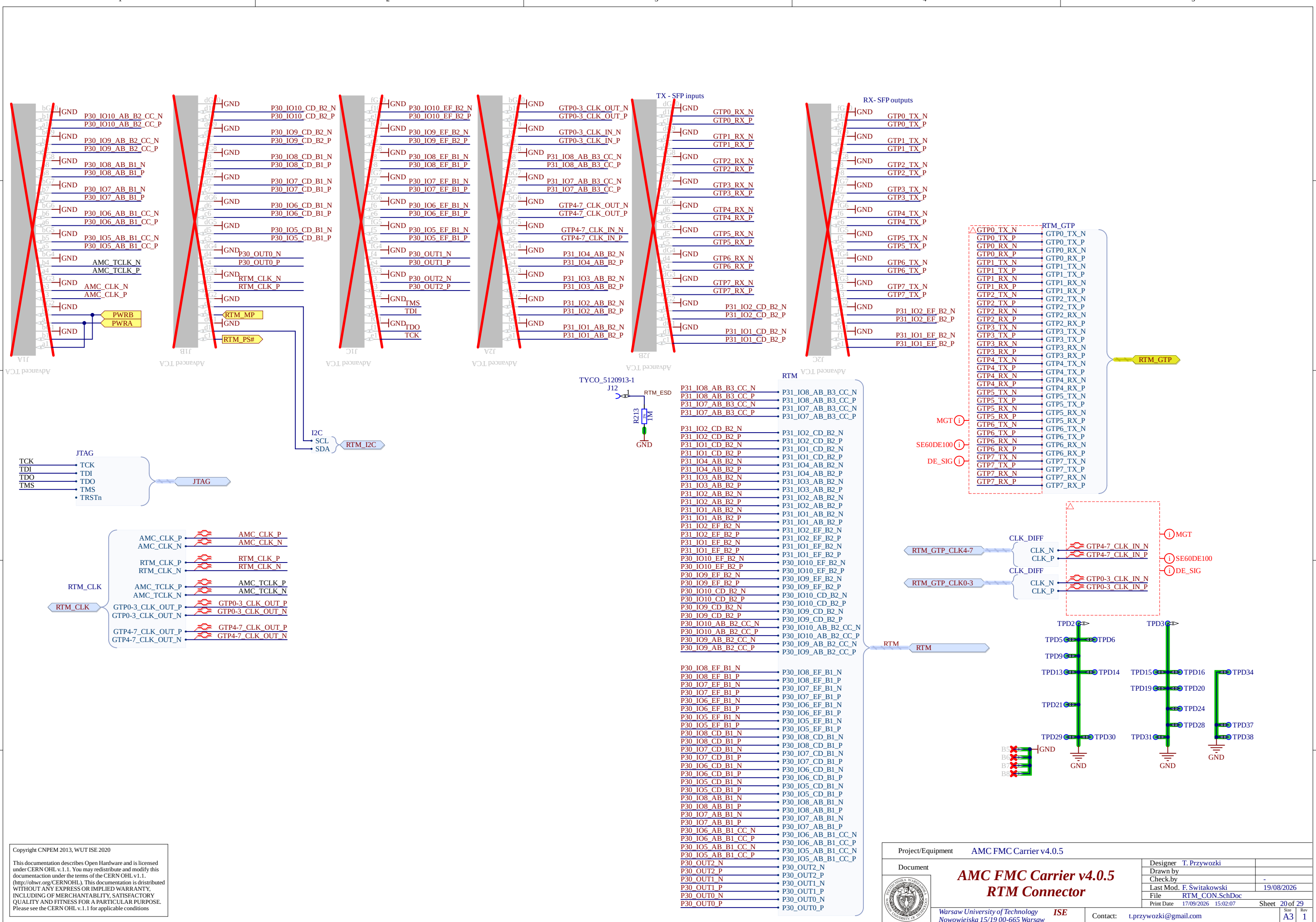
This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions



Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v1.1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1.1 for applicable conditions

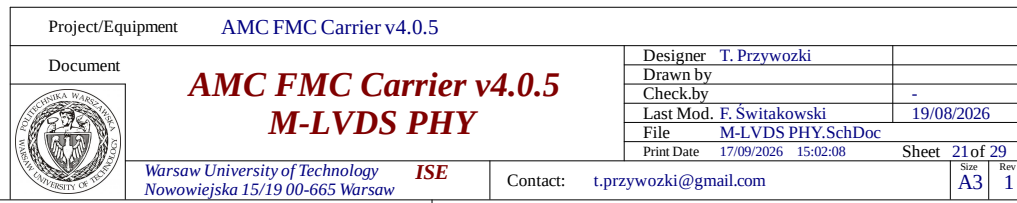
Project/Equipment		AMC FMC Carrier v4.0.5						
	Document AMC FMC Carrier v4.0.5 Thermometers		Designer	T. Przywozki				
			Drawn by	XX/XX/XXXX				
			Check.by	-				
			Last Mod.	F. Switakowski 19/08/2026				
			File	Thermometers.SchDoc				
			Print Date	17/09/2026 15:02:07				
			Sheet		19 of 29			
Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		ISE	Contact:	t.przywozki@gmail.com	Size	A4	Rev	1

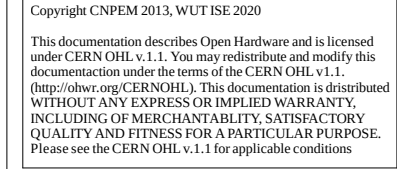


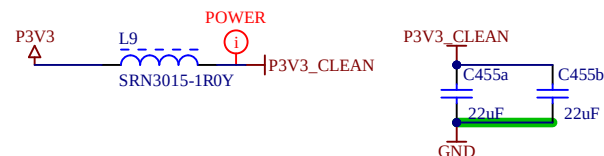
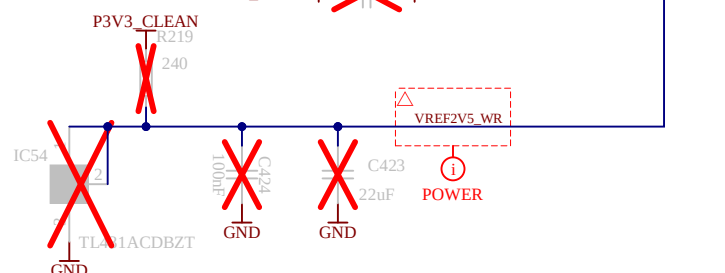
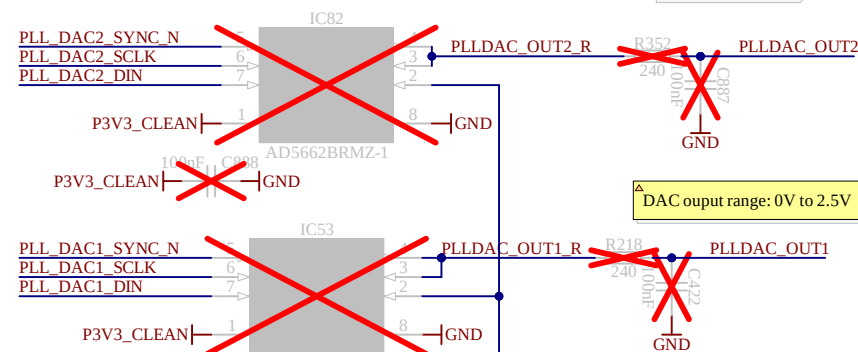
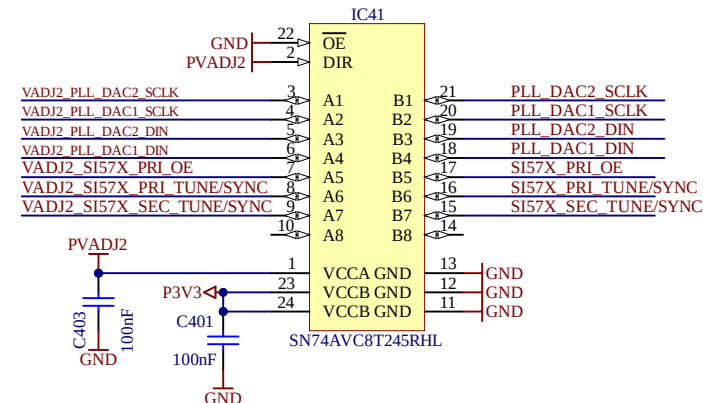
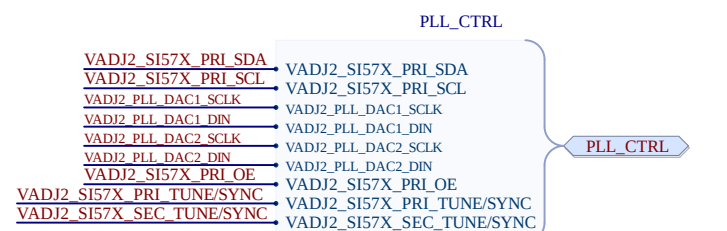
Copyright CNPEM 2013, WUT ISE 2020

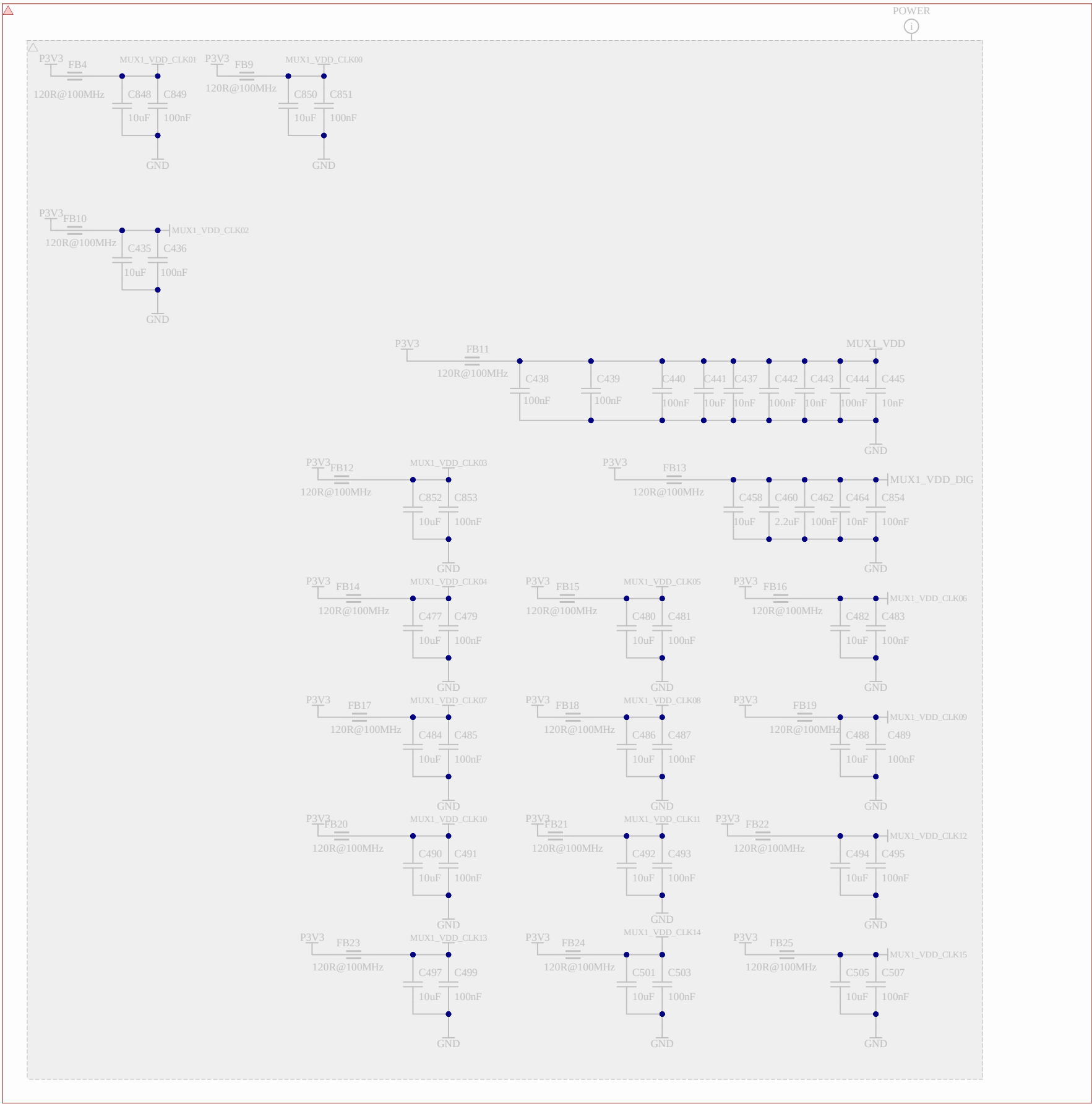
This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

Project/Equipment		AMC FMC Carrier v4.0.5	
Document		Designer T. Przywozki	
		Drawn by	-
		Check by	-
		Last Mod. F. Switkowski	19/08/2026
		File RTM_CON.SchDoc	
Warsaw University of Technology		ISE	Contact: t.przywozki@gmail.com
Nowowiejska 15/19 00-665 Warsaw		Print Date 17/09/2026 15:02:07	Sheet 20 of 29
		Size A3	Rev 1





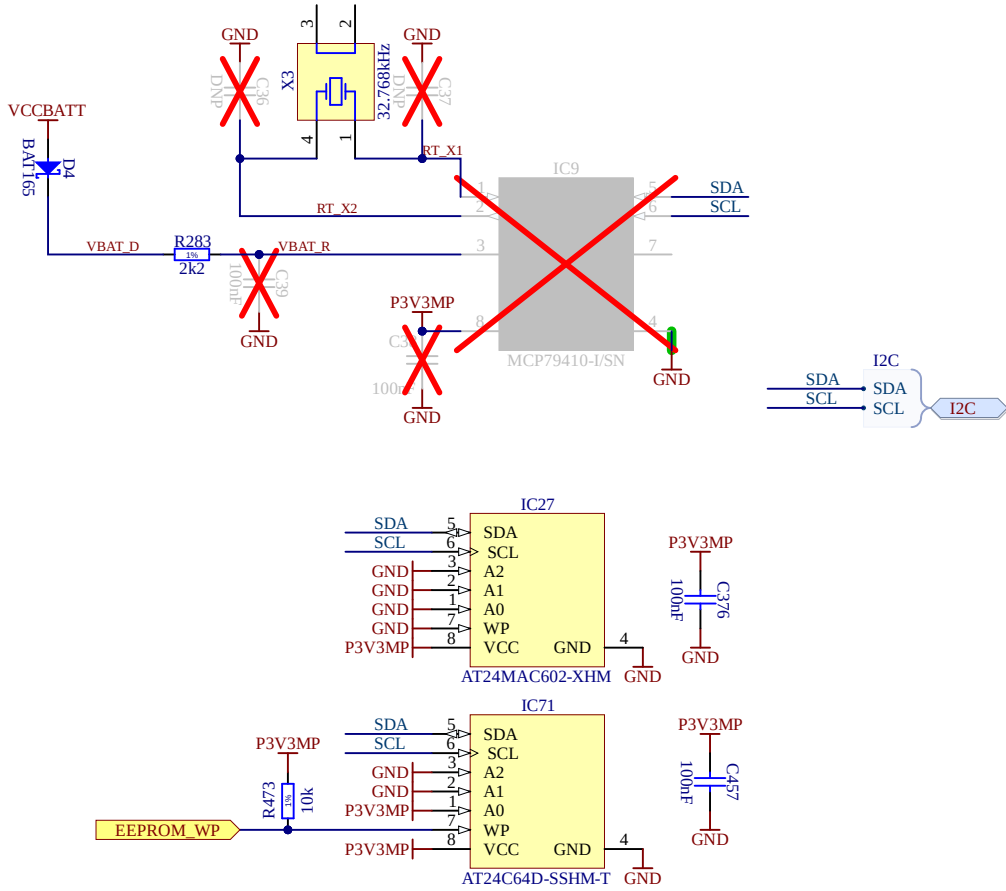




Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v.1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

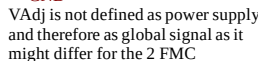
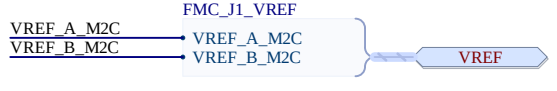
Project/Equipment		AMC FMC Carrier v4.0.5	
Document		Designer T. Przywozki	
		Drawn by	-
		Check by	-
		Last Mod. F. Świtakowski	19/08/2026
		File 8V54816A_SUPPLY.SchDoc	
Print Date 17/09/2026 15:02:09		Sheet 24 of 29	Rev 1
Warsaw University of Technology ISE		Contact: t.przywozki@gmail.com	Size A3

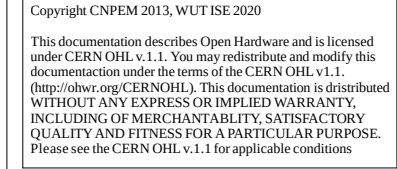


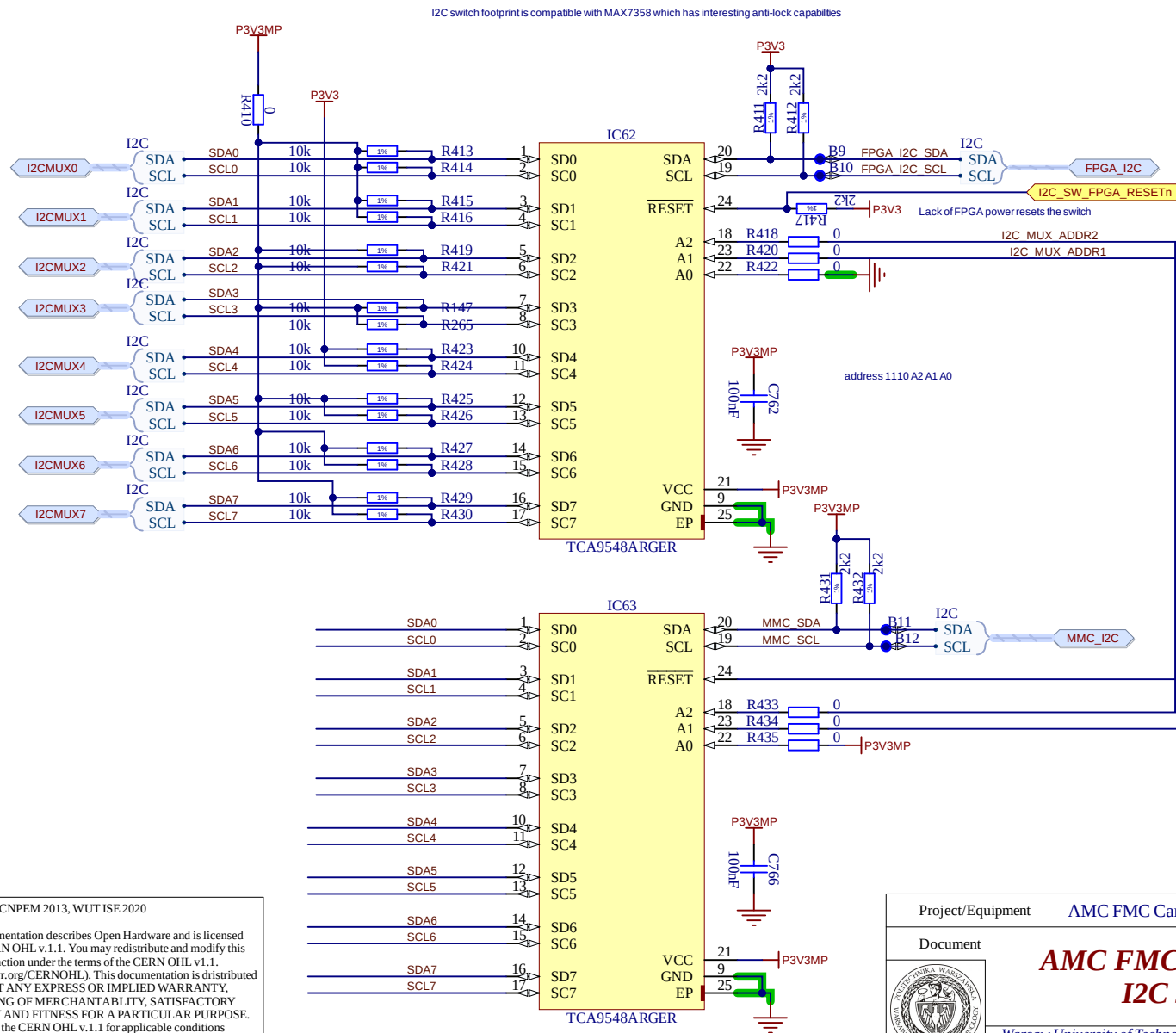
Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v1.1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1.1 for applicable conditions

Project/Equipment		AMC FMC Carrier v4.0.5					
	Document <i>AMC FMC Carrier v4.0.5</i> <i>RTCE & EEPROM</i>		Designer		T. Przywozki		
			Drawn by		XX/XX/XXXX		
			Check by		-		
			Last Mod.		F. Świtakowski	19/08/2026	
			File		RTCE.SchDoc		
			Print Date		17/09/2026 15:02:09	Sheet	26 of 29
			Contact:		t.przywozki@gmail.com		Size
Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		ISE		Rev	1		







I2C device address map The resources are available from TCA9548s SW: MMC I2C1, addr 1110 A2 A10 FPGA I2C and MMC I2C2, addr 1110 A2 A11 where A2=P2[28], A1=P2[29] default addr is 0x70 and 0x71	MUX Port 0	MAX6642 0x48
	Temp sensors	LM75: 0x4C LM75: 0x4D LM75: 0x4E LM75: 0x4F
	MUX Port 1	RTC EEPROM: 0x57
	RTCE	SRAM/RTCC reg: 0x58 EEPROM 2k: 0x50 EUI-ID: 0x59 EEPROM 64k: 0x51
	MUX Port 2	
	MUX Port 3	INA3221AIRGV: 0x40
	Power	INA3221AIRGV: 0x41 INA3221AIRGV: 0x42
	MUX Port 4	8V54816ANLG: 0x5B
	Clock switch	
	MUX Port 5	RTM devices
	RTM	
	MUX Port 6	EEPROM 0x52
	FMC2	
	MUX Port 7	EEPROM 0x50
	FMC1	

mux address 1110 A2 A1 0 from FPGA side 0x7x
 mux address 1110 A2 A1 1 from CPU side 0x7x

I2C MUX ADDR[2..1]

Copyright CNPEM 2013, WUT ISE 2020

This documentation describes Open Hardware and is licensed under CERN OHL v.1.1. You may redistribute and modify this documentation under the terms of the CERN OHL v1.1. (<http://ohwr.org/CERNOHL>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.1 for applicable conditions

Project/Equipment		AMC FMC Carrier v4.0.5	
	Document AMC FMC Carrier v4.0.5 I2C switches	Designer	T. Przywozki
		Drawn by	
		Check by	-
		Last Mod.	F. Świtkowski 19/08/2026
		File	I2C_MUX.SchDoc
		Print Date	17/09/2026 15:02:10
Warsaw University of Technology ISE Nowowiejska 15/19 00-665 Warsaw		Contact:	t.przywozki@gmail.com
		Sheet	29 of 29
		Size	A4
		Rev	1